Introduction to Probability and Statistics

4CM521

Full Lecture Notes



Illustration by Paul Warren. George E. P. Box (1919-2013): "All models are wrong, but some models are useful".

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1 Introduction

1.1 Contact Information

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1.2 Module Outline

This term will expose you to a range of common statistical methods. We are using **R** to perform analysis using a range of methods- understanding the underlying theory and being able to interpret **R** output are more important than being able to get a routine working. You are very strongly encouraged to take the time to see just how these functions work. There are worked solutions available for selected problems, but don't just type in commands- make sure you understand (and experiment with) the logic of the various procedures. In this module, for example, we will address the following topics:

- Organising and presenting data
- Descriptive statistics
- Sets and an introduction to probability
- Discrete and continuous random variables and their probability distributions
- Properties of random variables
- Discrete and continuous probability distributions
- Mean and variance of linear combinations of random variables
- Moment generating functions (MGF)
- Maximum Likelihood Estimation (MLE)

Do make sure you work through the questions in these lecture notes along with the questions in the tutorials. Also, you are reminded that you should spend as much time studying this topic in your own time as you do in lectures and seminars.

1.2.1 Learning Outcomes

The key learning outcomes of this term are:

1. Apply probabilistic methods to solve real-world problems
2. Demonstrate knowledge and understanding of a range of fundamental concepts in probability theory and applications.

1.2.2 Assessment Methods

- There will be 2 summative components, each carrying 50% of the overall mark.
    - Coursework

    - Exam
- Your progress will be monitored throughout the term and formative feedback will be provided.
- Informal feedback will be given to you during the seminars, with references to worksheets, as a group and on an individual basis.
- Comments will be given on the coursework and exam which will be handed back to you.

Please refer to the Introduction to Probability and Statistics Assessment Briefs for more detailed information about what is expected in each assessment. There are two assessment briefs: one for the coursework and another for the exam. These are available on Blackboard.

1.3 Teaching and Learning

Teaching Style: The lectures are designed to introduce new material. The tutorials will be computer guided (although myself and possibly another lecturer will be in attendance) to make sure you are clear what you need to take from the sessions.

Self-study: The advice given is that for every hour that you spend in classes, you should spend at least another hour working on this module in your own time. You will also need to spend a substantial amount of time working on coursework. In your self-study time you should do the following:

- Make sure you have a full set of notes and can answer the problems in the notes. Should you have to miss a class it is important that you catch up on what you miss from a friend.
- Read through the notes and make sure you understand them. Highlight anything which does not make sense to you and discuss it with a friend. You are also welcome to ask me about any part of the notes which you don't understand. Either catch me at the end of a lecture or during a tutorial class. You are also welcome to e-mail me or to come to find me in my office.
- Do some supplementary reading. Have a look at corresponding chapters in the recommended books or have a glance through other books in the library. The methods and techniques taught on this module are such important topics that there are many great web-resources - have a look at some of these. It is good to see how different people approach the same problem in different ways - it can help your understanding to no end. Also, you might find more interesting applications - let me know!
- Do finish off any problems in the notes which you don't get time to complete in class. Also, do make sure you complete the computer-based tutorials. After the first few weeks, these tutorials are designed to help consolidate your understanding of the material we cover in the lectures, so it is essential that you work through these exercises. By adopting a systematic approach, you will save yourself a great deal of time when you come to revise for the exam.

1.3.1 Useful textbooks

See below for some recommended reading linked to this module. They provide further discussions and examples.

- Rees, D. G. (2001). **Essential Statistics** (4th ed.), Chapman and Hall.
- Rizzo, L. M. (2019). **Statistical Computing with R** (2nd ed.), CRC Press
- Blitzstein, J.K., & Hwang, J. (2014). **Introduction to Probability** (1st ed.). Chapman and Hall/CRC.
- Mavrakakis, M.C., & Penzer, J. (2021). **Probability and Statistical Inference: From Basic Principles to Advanced Models** (1st ed.). Chapman and Hall/CRC.

2 Notation and Definitions

2.1 Learning Outcomes

At the end of this section, you must be able to:

- Understand and define fundamental symbols used in statistics.
- Develop the ability to read and write statistical notation fluently.
- Distinguish between population parameters (μ, σ^2) and sample statistics $(\bar{X}, s^2)$.

2.2 Introduction

Statistics, a powerful discipline within the realm of mathematics, plays a pivotal role in transforming raw data into meaningful insights and informed decisions. It provides a systematic framework for collecting, organising, analysing, interpreting, presenting, and drawing conclusions from data. In essence, statistics is the science of learning from data.

2.3 Definitions

2.3.1 Population[1]

A population is any entire collection of people, animals, plants or things from which we may collect data. It is the entire group we are interested in, which we wish to describe or draw conclusions about.

Example 2.1. Suppose an election is happening soon. We want to do a study looking at who people in the United Kingdom will vote. What is the population for this study?

Solution

All people who have registered to vote by the deadline.

2.3.2 Sample[2]

In practice, it is almost impossible to study the entire population. Thus, we need a smaller group which represents the population. We call this group a sample and define it as a group of units selected from the population. For each population there are many possible samples.

A sample is generally selected for study because the population is too large to study in its entirety. The sample should be representative of the population. This is often best achieved by random sampling.

[1] http://www.stats.gla.ac.uk/steps/glossary/basic_definitions.html
[2] http://www.stats.gla.ac.uk/steps/glossary/basic_definitions.html

It is important that the researcher/investigator has a complete and precise definition of the population before collecting the sample, including a description of the members to be included.

Example 2.2. All registered voters in Nottingham can be considered as a sample from the population for the study described in Example 2.1. Is this a good sample?

Solution

Not really. Nottingham doesn't represent the UK as a whole. (Nobody registered to vote in Nottingham will be voting SNP or Plaid Cymru!)

Example 2.3. An investigator is interested in the crime rate in different areas of Nottingham. To identify areas for their study, they use Nottingham postcodes (see http://en.wikipedia.org/wiki/NG_postcode_area). They decide to make their conclusion based on the 5 selected postcodes from all NG postcodes sorted alphabetically. Therefore,

Population of interest – Every postcode in the list

Sample – NG1, NG2 NG3, NG4 and NG5

Criticisms of the study – Doesn't represent the entire area. NG1-5 postcodes are all close to Nottingham city centre but the NG postcode region includes very different areas.

2.3.3 Parameters[3] and Statistics[4]

A parameter is usually an unknown value (and which therefore must be estimated), used to represent a certain population characteristic. For example, the population mean is a parameter that is often used to indicate the average value of a quantity.

A statistic is a quantity that is calculated from a sample of data. It is used to give information about unknown parameters in the corresponding population. For example, the average of the data in a sample is used to give information about the overall average in the population from which that sample was drawn.

A parameter is a fixed value within a population and does not vary. Considering the fact that it is possible to draw more than one sample from the same population, thus the value of a statistic will in general vary from sample to sample. In the population mean example, the average values in more than one sample, drawn from the same population, will not necessarily be equal.

Example 2.4. Suppose we want to know what percentage of people in the UK are over 180cm tall. We have a sample of 1000 people from the UK available.

Parameter of interest - The percentage of people in the population over 180cm tall in the UK.

[3] http://www.stats.gla.ac.uk/steps/glossary/basic_definitions.html#param
[4] http://www.stats.gla.ac.uk/steps/glossary/basic_definitions.html#stat

Statistic of interest - The percentage of people over 180cm in the sample is a statistic that we could use to estimate our parameter of interest.

2.3.4 Taking a Sample

Clearly a sample with a reasonable size should be as representative of the population from which it is drawn as possible. There are many, many examples in the newspapers and magazines of reports and conclusions based on unrepresentative and/or unreasonable small samples. You have already seen such samples in Examples 2.2 and 2.3.

Example 2.5. Suppose we want to do a study on what people's favourite sport is in the UK. Person A stands outside a rugby stadium on match day and ask 10 people to name their favourite sport. Person B stands in Nottingham City Centre and asks 1000 people to name their favourite sport. Whose study is better? Is either perfect?

Solution

Person B's study is clearly better as they have a larger sample from people that are more representative of the population as a whole. It's still not perfect though. People in Nottingham alone are probably not representative of the entire UK.

These examples highlight how important it is to select the sample to give an accurate reflection of the characteristics of the whole population. There are possible ways of choosing a sample from a given population. These ways are known as the sampling methods in statistics and the most common ones are simple random sampling, cluster sampling, systematic sampling and quota sampling. Here we only describe simple random sampling.

Example 2.6. Random Sampling

Do it in **R**:

```
#To sample 3 elements from a vector 1:10 without replacement:
x<- sample(1:10, 3)
x

## [1] 6 3 7

#To sample 3 elements from a vector 1:10 with replacement:
y<- sample(1:10, 3, replace = TRUE)
y

## [1] 4 8 7

#Random sample with weighted probabilities
probabilities <- c(0.1, 0.1, 0.2, 0.2, 0.05, 0.05, 0.1, 0.05, 0.05, 0.1)
z<- sample(1:10, 3, prob = probabilities)
z

## [1] 4 8 3
```

2.4 Types of Data

The data can be classified into different categories. We will see that different analytical methods are appropriate for different types of data. The data can be categorised into two main types:

- Qualitative data, also known as categorical data, these types of data are non-numerical and can be either nominal or ordinal
- Quantitative data, these types of data are numerical and can be either discrete or continuous.

2.4.1 Nominal data

Nominal data consist of simple labels such as male/female/non-binary, employed/unemployed, geographical area such as American/Afghan/Northern Indian/Oriental. It does not matter whether we write yes/no or no/yes. They can take on numerical values (such as “0” indicating no and “1” indicating yes), but those numbers don’t have mathematical meaning.

Examples: Gender, nationality, university attended

2.4.2 Ordinal data

Ordinal data have a natural or inherent ordering. For example, the response to a question asking respondents to rate the service of the retailer from 1 to 5 where 1 is “poor” and 5 is “excellent” is an example where the categories are ordinal.

Data such as age or salary may also be ordinal if they have been categorised into groups for the purposes of data collection. (Under 16, 16-25, 26-40, 41-64, 65+)

Examples: Ratings out of 10, categorised ages, university grade

2.4.3 Discrete data

Discrete data usually arises from counting something and consists of whole numbers. The data can only take specific separate values such as 0, 1, 2 etc. For example, the number of children under 18 in a family is discrete.

As with all quantitative data, the data have mathematical properties. A respondent who has 4 children under 18 has twice as many as one who has 2.

Examples: Shoe size, number of exams taken, goals scored

2.4.4 Continuous data[5]

Continuous data can only be described using intervals on the real number line. They usually arise from measuring something such as time, weight, height, or length. Continuous data will always be rounded. The degree of accuracy used depends on

[5] The precision of measurement affects what is recorded in practice; for example, someone’s height may be 178.375621 cm to six decimal places, but it may only be recorded as 178 cm.

the situation. For example, someone's height is a continuous variable. The respondents may report their height to the nearest whole number or to the nearest ½ whole number.

If continuous data are rounded, then they will look like discrete data. Thus, ask yourself if this is a measure that has been rounded. If it is then the data are continuous.

Examples: Height, weight, distance travelled

3 Organising and Presenting Data

3.1 Learning Outcomes

At the end of this section, you must be able to:

- Choose and create suitable graphical representations, such as histograms, bar charts, and scatterplots, for different types of data.
- Assess the quality of data presentations, identifying potential errors, inconsistencies, or misleading representations.

3.2 Introduction

So far, we have discussed the different types of data available to us and have looked at ways of summarising data numerically. We will now look at how we can visualise data using plots and charts to convey our information to a reader. Note that the following general rules should always be applied when illustrating any type of plot or chart.

- Label the axes clearly
- Include scales on the axes and the units used
- Title the chart (give a caption as well if necessary)
- Present the information clearly and unambiguously
- Include the source of data it possible

All of the following plots can be produced in **R** and we will look to do this in class.

3.3 Bar Chart

There are several variations on the bar chart theme which will be illustrated in this section. We start with the following example.

Example 3.1. A simple bar chart.

The following data is from an experimental study of recidivism of 432 male prisoners who were observed for a year after being released from prison[6].

Tabel 3.1 shows out of total 432 prisoners, 114 prisoners convicted a crime after having been previously convicted within a one year.

Table 3.1. 432 prisoners who were observed for one year after being released from prison.

		Total
Recidivist	No	318
	Yes	114

[6] Rossi, P. H., R. A. Berk & K. J. Lenihan. 1980. Money, Work and Crime: Some Experimental Results. New York: Academic Press.

The bar chart in 3.1 is an example of a simple bar chart. These types of graphs are often used for presenting *qualitative* data sets. The height of the bar corresponds to the count. It can be seen that the majority of prisoners do not reoffend within a year after being released.

Do it in **R**:

```
#Sampledata
categories<-c("No","Yes")
values<-c(318,114)

#Createabarchart
barplot(values,names.arg=categories,main="Recidivist",
      xlab = "Categories", ylab = "Values", col = "blue", ylim = c(0,350))
```

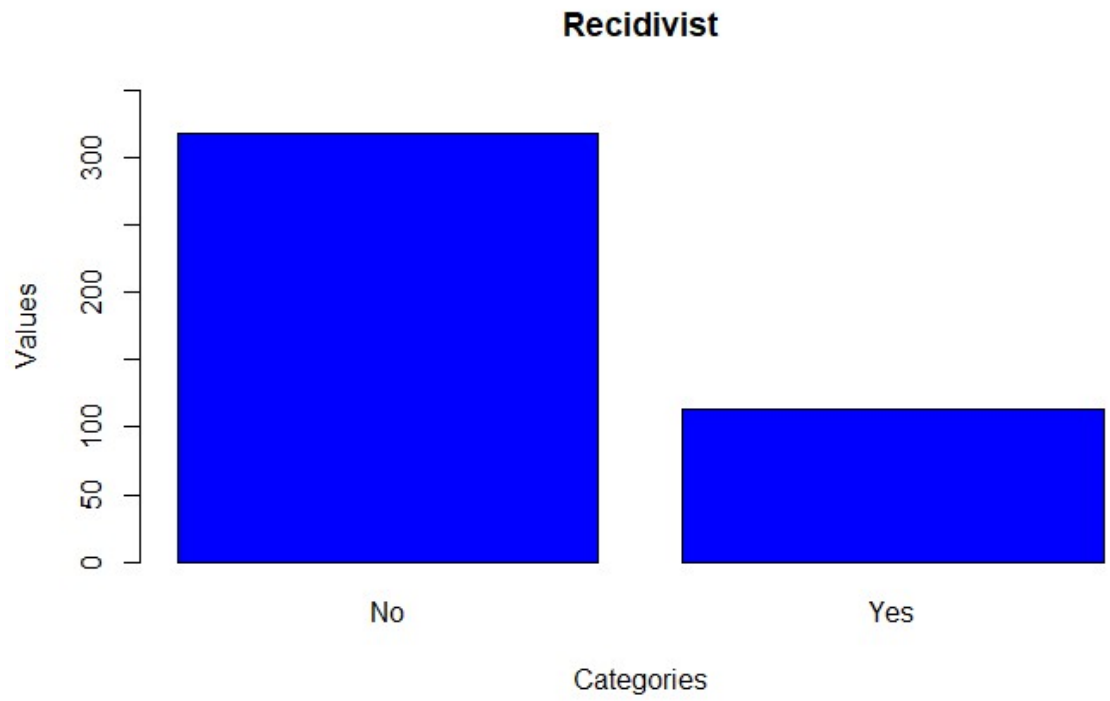


Figure 3.1. A simple bar chart corresponding to the statistics presented in Table 3.1.

Note that we can assign numbers to the categorical data in Example 3.1, e.g. "0" indicating No and "1" indicating Yes), however we cannot observe numbers between 0 and 1, 1 and 2 etc., and this is shown by the gaps between the bars. This distinguishes the plot from the histogram used for continuous data that we shall see later.

Advantage – Simple, easy to read.

Disadvantage – Doesn't contain much information.

Example 3.2. A composite bar chart.

Table 3.2 compares the number of prisoners who had full-time work experience with prisoners who did not have full time work experience prior to incarceration grouped by whether they repeated a crime or not.

Table 3.2. 432 prisoners (with and without full time work experience) who were observed for one year after being released from prison.

		Full time employed		
		No	Yes	Total
Recidivist	No	123	195	318
	Yes	62	52	114
	Total	185	247	432

Do it in **R**:

```
# Sample data
categories <- c("No", "Yes")
values1 <- c(123, 195)
values2 <- c(62, 52)

# Create a matrix with the data
data_matrix <- matrix(c(values1, values2), nrow = 2)

# Define colors for the bars
bar_colors <- c("blue", "red")

# Create the bar chart
barplot(data_matrix, beside = FALSE, names.arg = categories,
        main = "", xlab = "Recidivist", ylab = "Values",
        col = bar_colors, ylim = c(0, 350))

# Add a legend
legend("topright", legend = c("Full time employed", "Not full time employe
d"), fill = bar_colors)
```

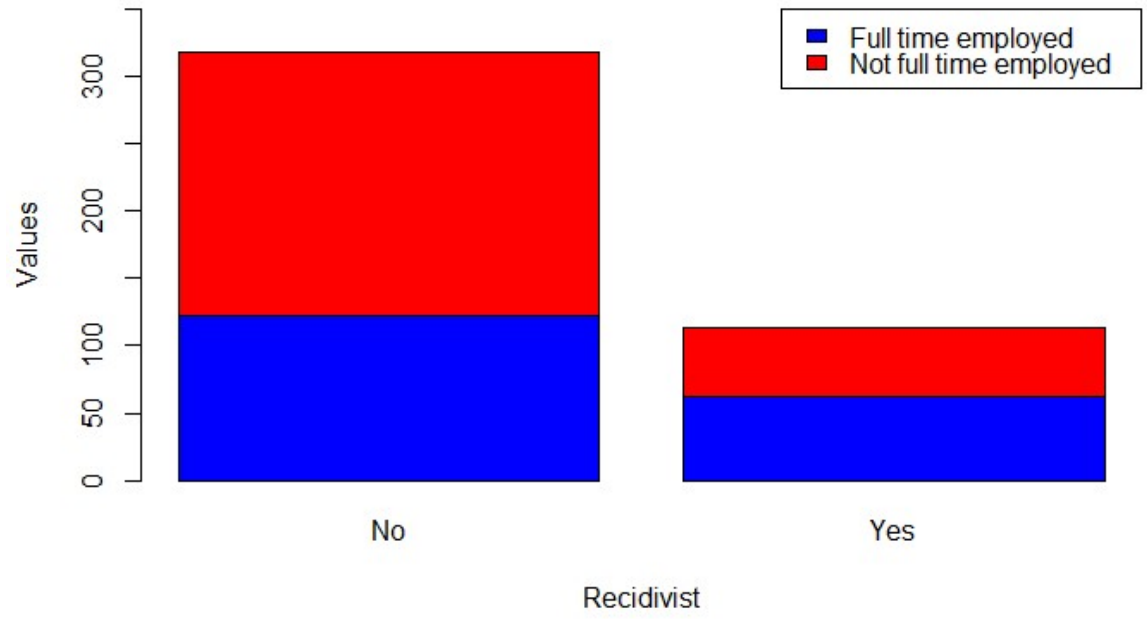


Figure 3.2. A composite bar chart using the data given in Table 3.2.

Advantage – Contains more information than the simple bar plot in the same amount of space.

Disadvantage – Can be difficult to compare some of blocks of the same shading and can't compare proportions easily.

Example 3.3. A multiple bar chart.

Using the data in Table 3.2 we can create a multiple bar chart as shown in Figure 3.3. We can see that prisoners with full time work experience are slightly less likely to repeat a crime compared to unemployed prisoners.

Do it in **R**:

```r
# Sample data
categories <- c("No", "Yes")
values1 <- c(123, 195)
values2 <- c(62, 52)

# Create a matrix with the data
data_matrix <- matrix(c(values1, values2), nrow = 2)

# Define colors for the bars
bar_colors <- c("blue", "red")

# Create the bar chart
barplot(data_matrix, beside = FALSE, names.arg = categories,
        main = "", xlab = "Recidivist", ylab = "Values",
        col = bar_colors, ylim = c(0, 350))
```

```
# Add a legend
legend("topright", legend = c("Full time employed", "Not full time employe
d"), fill = bar_colors)
```

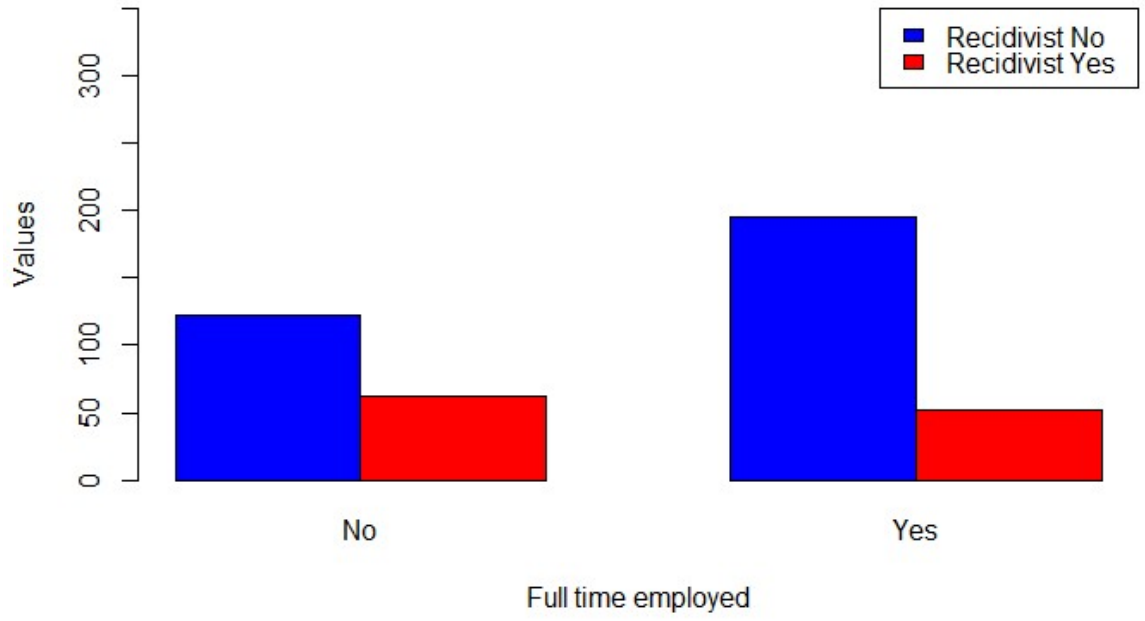


Figure 3.3. A multiple bar chart using the data given in Table 3.2.

Advantage – Can now compare the individual blocks.

Disadvantage – Takes up more space on the plot and still can't compare proportions easily.

3.4 Pie Chart

In this chart a circle is divided into segments which represent the quantity being charted. The angle each segment makes at the centre of the circle is obtained by multiplying 360 degrees by the appropriate proportion.

Example 3.4. Pie Chart.

Table 3.3 categorises the prisoners into 5 categories based on their education. We show how the calculations for a pie chart can be done. See Table 3.3 and 3.4.

Tabel 3.3. Prisoners under the study grouped by their education level.

Education	Frequency
Grade 6 or less	24
Grades 6 through 9	239
Grades 10 and 11	119

Grade 12	39
Some post-secondary	11

Tabel 3.4. Prisoners frequencies with corresponding proportions and angles.

Education	Frequency	Proportion	Angle
Grade 6 or less	24	24/432=0.06	0.06 × 360 = 20
Grades 6 through 9	239	239/432=0.55	0.55 × 360 = 199
Grades 10 and 11	119	119/43=0.28	0.28 × 360 = 99
Grade 12	39	39/432=0.09	0.09 × 360 = 33
Some post-secondary	11	11/432=0.03	0.03 × 360 = 9
Total	432	432/432=1	360

Do it in **R**:

```
# Data
education_data <- c(24, 239, 119, 39, 11)
categories <- c("Grade 6 or less", "Grades 6 through 9", "Grades 10 and 11
", "Grade 12", "Some post-secondary")

# Calculate percentages
percentages <- round(education_data / sum(education_data) * 100, 1)

# Create a pie chart with percentages
pie(education_data, labels = paste(categories, "\n", percentages, "%"), ma
in = "")
```

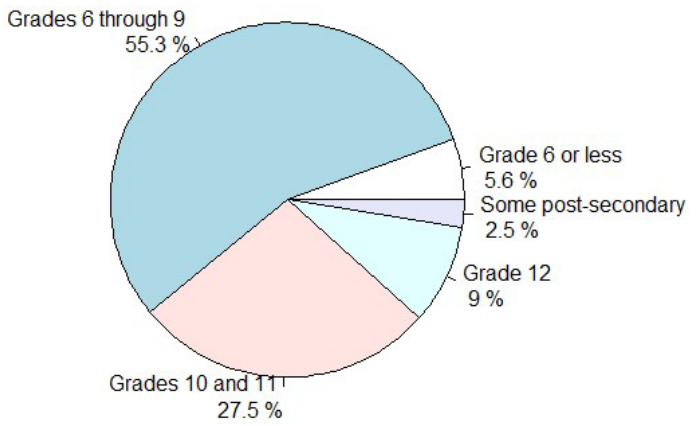


Figure 3.4. A pie chart using the data given in Table 3.3.

Advantage – Look very pleasing to the eye, hence their widespread use

Disadvantage – The human eye is MUCH better at comparing heights than angles so pie charts are often harder to interpret. They are also easy to manipulate by using bolder colours to make certain segments look bigger. A bar chart would be much better for representing these data. Never use a pie chart!

3.5 Histograms

Histograms, as in Figure 3.5, can also be used, particularly for larger data sets. It can be seen that majority of prisoners were 18 to 20 years old (25 prisoners). The area of each bar is proportional to frequency; as the bars in this figure are of equal width, this means that height is also proportional to frequency.

Do it in **R**:

```
#hypothetical (hence sampling)
probabilities <- c(0.1, 0.09, 0.15, 0.1, 0.05, 0.05, 0.05, 0.05, 0.05, 0.0
6, 0.05, 0.04, 0.03, 0.03, 0.02, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01,
0.001)
x<- sample(18:40, size = 432, replace = TRUE, prob = probabilities)
hist(x, xlim=c(15,50), ylim = c(0,200), xlab = "Age", ylab = "Frequency",
main = "Prisoner Age Distribution")
```

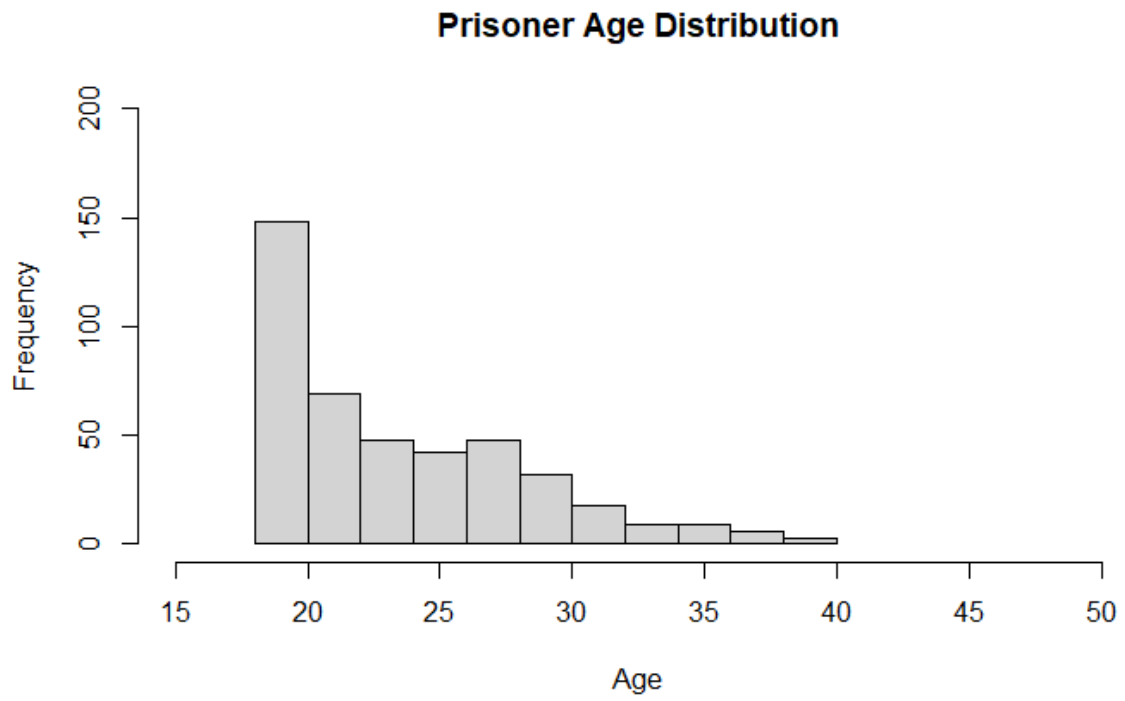


Figure 3.5. Histogram of age of prisoners.

4 Descriptive Statistics

4.1 Learning Outcomes

At the end of this section, you must be able to:

- Calculate and interpret measures of central tendency and dispersion.

4.2 Introduction

We will learn how to summarise quantitative data numerically into few statistics. In particular, we look at measures of location and measures of variation.

4.3 Measures of Location

These statistics are designed to give a few *"typical"* values in our data to indicate what we can expect from the population overall.

4.3.1 Minimum

$\mathbb{DF}$: The minimum is the smallest value in a data set. It is often useful to put data in ascending order (smallest to largest) when studying it.

Example 4.1. Let X represent the following values: $\{1, 2, 3, 6, 7, 9, 11\}$. What is the minimum?

Solution

From this ordered listing, it is immediate that the minimum of X is 1.

Do it in **R**:

```
x<-c(1,2,3,6,7,9,11)
min(x)
```

```
## [1] 1
```

4.3.2 Maximum

$\mathbb{DF}$: The maximum is the largest value in a data set.

From the above ordered list Example 4.1, the maximum of X is 11.

Look at Example 3.2 and find the maximum number of weeks until reoffending.

Do it in **R**:

```
x<-c(1,2,3,6,7,9,11)
max(x)
```

```
## [1] 11
```

4.3.3 Median

$\mathbb{DF}$: The median is the middle value in the sense that half the data are above it, and half the data are below it.

Note, if the number of data points is an odd number, the median is the middle value, e.g., 6 for data set X (4th observation in the ordered list). If there are an even number of data, the median is half way between the two middle values.

Example 4.2. Let Y represent the following values $\{12, 14, 14, 16, 17, 17, 20, 20\}$. What is the median?

Solution

Since there are 8 values then $(4^{\text{th}} + 5^{\text{th}})/2 = 4.5^{\text{th}}$. Thus, $(16 + 17)/2 = 16.5$. When finding the median, make sure the data are ordered ascending, and each value has been listed as often as it occurs. The median is perhaps the best indicator of where the centre of the data is.

Do it in **R**:

```
x<-c(12,14,14,16,17,17,20,20)
median(x)

## [1] 16.5
```

4.3.4 Mean

$\mathbb{DF}$: The mean of a given data set is the sum of the data points divided by the number of data points. This is mathematically defined as:

$$\bar{X} = \frac{\sum_i^n x_i}{n} = \frac{\text{Sum of the observations}}{\text{Number of observations}}$$

where $\bar{X}$ (pronounced 'x bar') denotes the mean, $\sum$ (the Greek letter capital 'sigma') is the mathematical notation for adding up the terms that follow, $i = 1, \ldots n$ are iterations in the summation where n is the total number of observations.

Example 4.3. The data set consist of the numbers $X = (1,4,7,3,9,2)$. What is the mean?

Solution

$$\sum_{\text{all } x} x = 1 + 4 + 7 + 3 + 9 + 2$$

More formally, the mean is defined and obtained as

$$\bar{x} = \frac{\sum_1^6 x_i}{6} = \frac{1 + 4 + 7 + 3 + 9 + 2}{6} = \frac{26}{6} = 4.33 \ldots$$

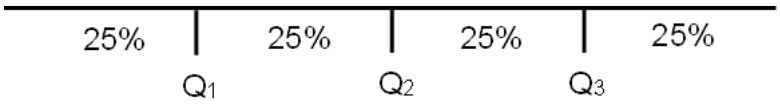
Do it in **R**:

```
x<-c(1, 4, 7, 3, 9, 2)
mean(x)
```

```
## [1] 4.333333
```

4.3.5 Quartiles

$\mathbb{DF}$: The quartiles of a set of data divide the data into four equal parts and they are denoted as Q_1, Q_2 (the median) and Q_3.



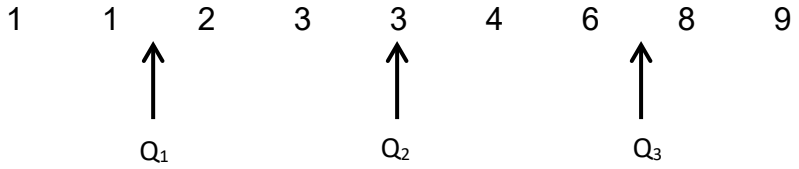
In order to calculate quartiles, we follow the following steps:

1. Order the data, from smallest to the largest
2. The first quartile is the point taken to be $\frac{n+1}{4}$ of the way along the ordered data set. (Remember n represents the number of data points)
3. The second quartile is the point taken to be $\frac{2(n+1)}{4}$ (or half) of the way along the ordered data set. This is also the median.
4. The third quartile is the point taken to be $\frac{3(n+1)}{4}$ of the way along the ordered data set.

Example 4.4. Find the quartiles for $x = (1, 1, 4, 6, 3, 3, 9, 8, 2)$.

Solution

1. Order the data from the smallest to the largest



2. $n = 9$, thus $(n+1)/4 = (9+1)/4 = 2.5$ that is Q_1 is halfway between the second and third largest observation. Hence:

$$Q_1 = 1 + \frac{(2-1)}{2} = 1 + 0.5 = 1.5$$

3. Q_2 is the median and is the $2 \times (n+1)/4 = 2 \times (9+1)/4 = 5^{\text{th}}$ largest observation. That is

$$Q_2 = 3.$$

4. Similarly, $3 \times (n+1)/4 = 3 \times (9+1)/4 = 7.5$, therefore Q_3 is halfway between the seventh and eighth largest observation. Hence:

$$Q_3 = 6 + \frac{(8-6)}{2} = 6 + 1 = 7$$

4.4 Measures of Variation

Measure of variation is a measure that describes how spread out or scattered a set of data is. It is also known as measures of dispersion or measures of spread. You can find out how the data are relatively close or far apart spread out. For instance, a low measure of variation will verify that values are relatively close together. There are different ways of measuring variation: range, interquartile range, variance and standard deviation.

4.4.1 Range

$\mathbb{DF}$: The range of a data set is the difference between the maximum and minimum values in the set.

Example 4.5. Find the range of the data set: $11, 10, 8, 4, 6, 7, 11, 6, 11, 7$

Solution

$$\text{Maximum} = 11$$

$$\text{Minimum} = 4$$

$$\text{Range} = \text{Maximum} - \text{Minimum} = 11 - 4 = 7$$

Finding the range is easy to compute. Range depends only on the highest and lowest values. However, it may not be as useful as other measures of variation.

Do it in **R**:

```
x<-c(11, 10, 8, 4, 6, 7, 11, 6, 11, 7)
range(x)

## [1]  4 11

range<- max(x)-min(x)
range

## [1] 7
```

4.4.2 Interquartile Range

$\mathbb{DF}$: In descriptive statistics, the interquartile range (IQR), also called the midspread or middle fifty, is a measure of statistical dispersion, being equal to the difference between the upper and lower quartiles, i.e. $IQR = Q_3 - Q_1$.

Example 4.6. Find the IQR of the data set: 11, 10, 8, 4, 6, 7, 11, 6, 11, 7

Solution

Order = $4, 6, 6, 7, 7, 8, 10, 11, 11, 11$

Since $n = 10$, then $(3 \times (10 + 1))/4 = 8.25$ then Q_3 is as follows:

$$Q_3 = 11 + \frac{(11 - 11)}{4} = 11 + 0 = 11$$

also, $(10 + 1)/4 = 2.75$, then Q_1 is as follows:

$$Q_1 = 6 + \frac{3 \times (6 - 6)}{4} = 6 + 0 = 6$$

The IQR is given by:

$$IQR = Q_3 - Q_1 = 11 - 6 = 5$$

4.4.3 Variance and Standard Deviation

$\mathbb{DF}$: The variance is a measure of how far a set of numbers is spread out. A measure of the dispersion of a set of data points around their mean value.

Here is how to calculate it:

1. Subtract the mean from each value in the data. This gives you a measure of the distance of each value from the mean.
2. Square each of these distances (so that they are all positive values) and add all of the squares together.
3. Divide the sum of the squares by the number of values in the data set minus 1.

The standard deviation (σ) is simply the square root of the variance, which in turn is:

$$\text{Variance} = \sum_{i=1}^{n} \frac{(x_i - \bar{X})^2}{n - 1}$$

For our numbers (1, 3, 5) with a mean of 3 this gives:

$$\sigma^2 = \text{Variance} = \frac{(1 - 3)^2 + (3 - 3)^2 + (5 - 3)^2}{3 - 1} = 4$$

So, the standard deviation, the square root of this is as follows:

$$\text{standard deviation} = \sigma = \sqrt{4} = 2$$

The variance is, roughly, the average squared distance from the mean. The standard deviation has the property that, for 'well-behaved' and large data sets about 95% of the data should lie within two standard deviations of the mean.

Do it in **R**:

```
x<-c(1, 3, 5)
var(x)

## [1] 4

sd(x)
```

```
## [1] 2
```

5 Sets, Events and Probability

5.1 Learning Outcomes

At the end of this section, you must be able to:

- Define and understand the concept of sets, including notation and basic operations (union, intersection, complement).
- Define probability as a measure of likelihood and understand its range (0 to 1).
- Define and calculate conditional probabilities.
- Define and calculate the Law of Total Probabilities
- Understand the derivation and principles behind Bayes' theorem (and apply it to problems).

5.2 Introduction

In the realm of probability theory, understanding sets and events serves as the cornerstone for unraveling uncertainty and predicting outcomes. A set, a collection of distinct elements, lays the foundation for defining events, which are subsets representing specific outcomes within a larger context.

5.3 The algebra of Sets

If we think for a few moments about our experience of daily life, we will soon have to concede that many of the things that happen to us are uncertain. We are not too sure exactly how long this lecture will last even though it is timetabled for 50 minutes. Before we go to bed at night, we cannot say exactly what the weather will be like when we wake up in the morning. The list is endless. If we are truthful with ourselves and think long and hard about our lives, then we are liable to conclude that there are very few things that we can say about ourselves with certainty.

In statistics we are concerned with presentation and interpretation of chance outcomes that occur in a planned study or experiment. Statisticians use word experiment to describe any process that generates a set of data. We shall refer to any recording of information, whether it is numerical or categorical (names, YES/NO) as observations.

$\mathbb{DF}$: The set of all possible outcomes of a statistical experiment is called the sample space.

If we denote it by Ω then $\Omega = \{H, T\}$ for tossing the coin and $\Omega = \{1, 2, 3, 4, 5, 6\}$ for numbers that appear on top face of the experiment of tossing a die.

Example 5.1. Three items are selected at random from a manufacturing process. Each item is inspected and classified as defective (D) or non-defective (N).

$$\Omega = \{DDD, DDN, DND, \dots, NNN\}, |\Omega| = 8$$

Sample space with a large or infinite number of sample points are best described by a statement or rule.

$$\Omega = \{x|\ x\ is\ a\ city\ with\ a\ population\ over\ 1\ million\}$$

$$\Omega = \{x, y|\ x^2 + y^2 \leq 4\}$$

$\mathbb{DF}$: To every point in the sample space, we assign the probability such that the sum of all probabilities is 1.

For any given experiment we may be interested in the occurrence of certain events rather than in a specific outcome/element of sample space.

In Example 5.1 we may be interested in event B when the number is of defectives ≥ 2. This will occur if the outcome is an element of the subset.

$$B = \{DDD, DDN, DND, NDD\}, B \subset \Omega$$

$\mathbb{DF}$: An event is a subset of a sample space. For example, $B = \{x, y|\ x^2 + y^2 \leq 1\}$ is a subset of sample space $\Omega = \{x, y|\ x^2 + y^2 \leq 4\}$.

$\mathbb{DF}$: The complement of an event A with respect to sample space Ω is the subset of all elements of Ω that are not in A. For example, if R be the event that the red card is selected from the ordinary deck of 52 playing cards then $\bar{R}$ is the event that the card selected is not red but black.

5.4 Set Theory

Set theory notation offers a convenient way of explaining certain operations with the events and some of the laws of probability theory.

Before proceeding with the elementary laws of probability we first consider certain definitions from set theory that will appear in the laws.

Suppose we consider two events from the sample space of a given situation. Let us denote two events by the letters A and B. There events are represented by the circles in the diagrams below.

5.4.1 Union

$A \cup B$ denotes the union of A and B (either A occurs, or B occurs or both A and B occur). This is represented in Figure 5.1. The circles in the diagram overlap so it is possible for both events to occur at the same time. The events are therefore referred to as being non-mutually exclusive.

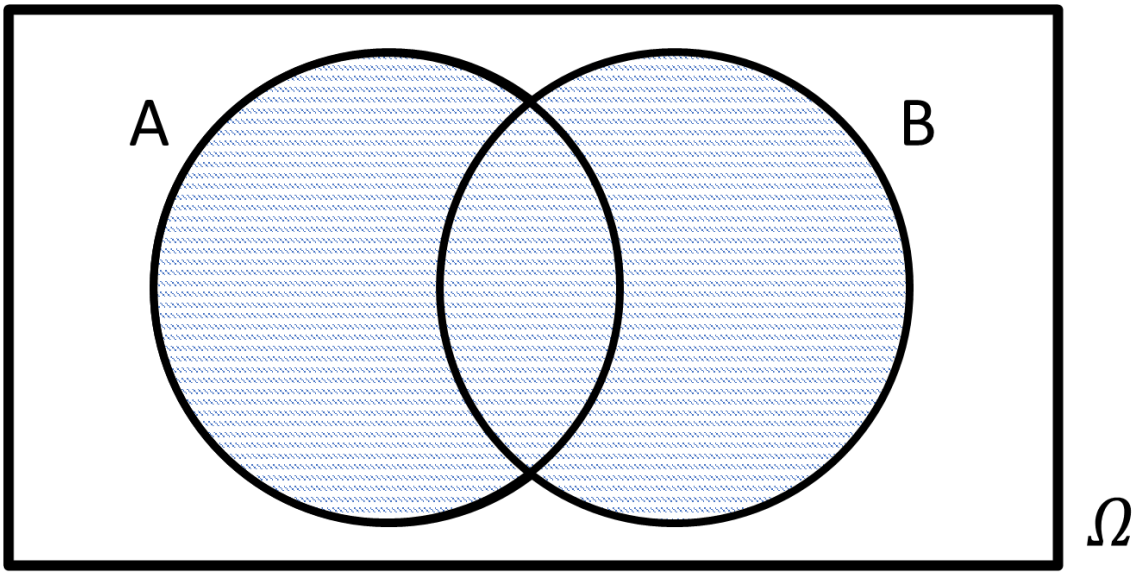


Figure 5.1. A representation of the union between A and B.

$\overline{A \cup B}$ denotes the complement of $A \cup B$ (neither A or B occurs). This is represented in Figure 5.2.

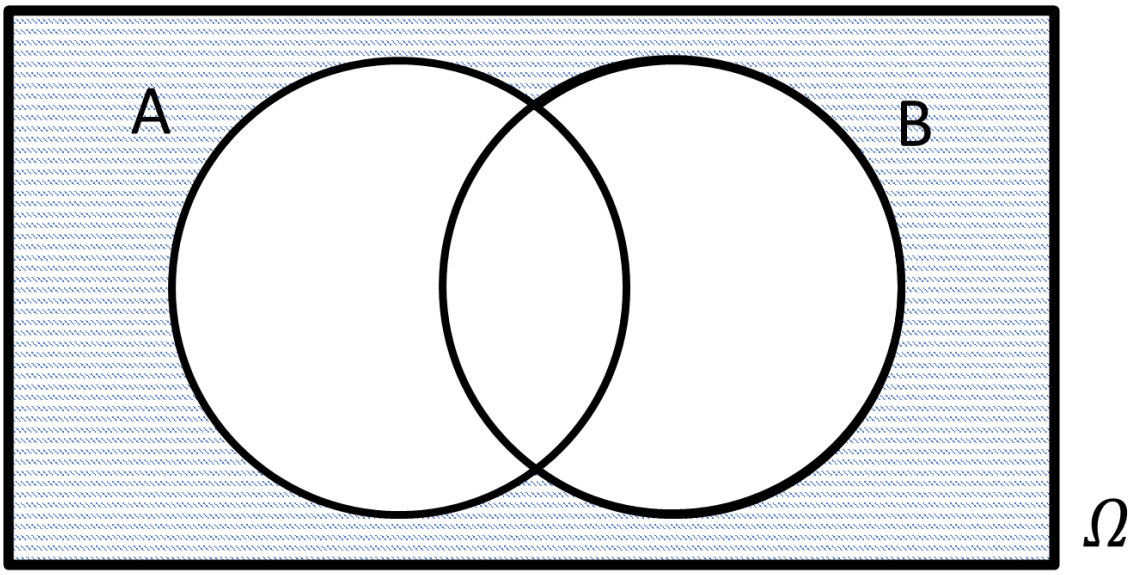


Figure 5.2. A representation of the complement of $A \cup B$. This is denoted as $\overline{A \cup B}$.

5.4.2 Intersection

$A \cap B$ denotes the intersection of A and B (both A and B occur). This is represented in Figure 5.3.

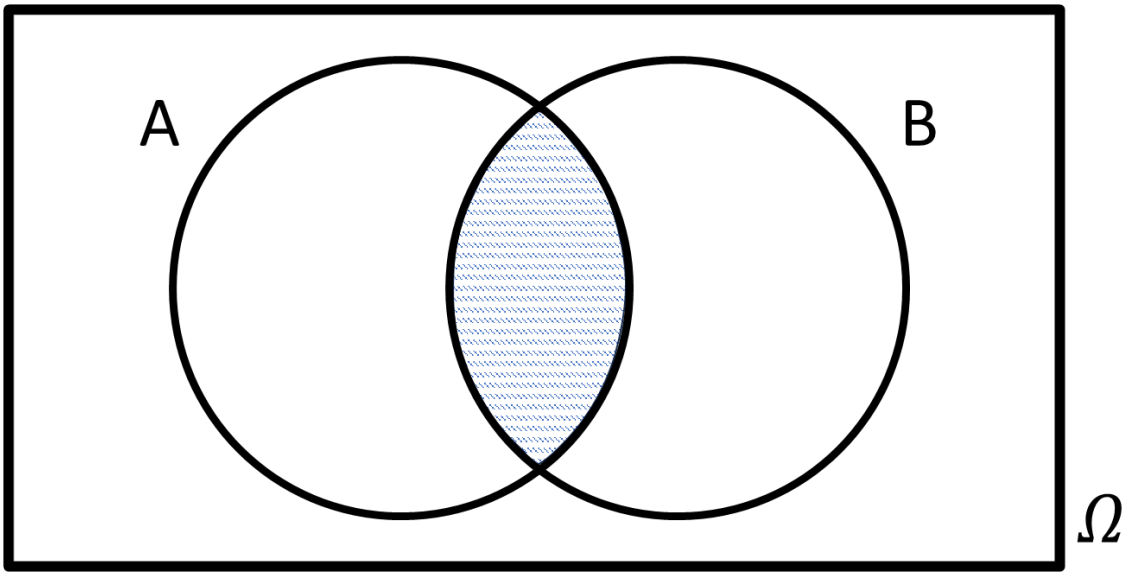


Figure 5.3. A representation of the intersection of A and B.

The complement of $A \cap B$ is denoted at $\overline{A \cap B}$. This is represented in Figure 5.4.

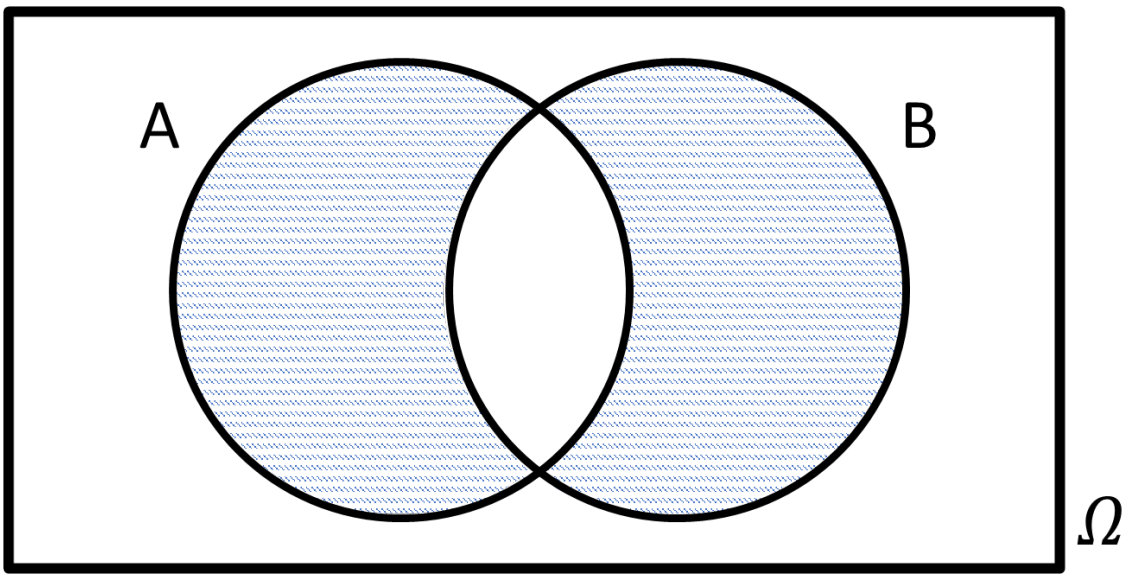


Figure 5.4. A representation of the complement of $A \cap B$. This is denoted as $\overline{A \cap B}$.

5.4.3 Mutually Exclusive Events

The intersection $A \cap B$ is the empty set, i.e. the two circles do not overlap. The two events cannot possibly occur together and are referred to as being mutually exclusive. This is denoted as $A \cap B = \phi$ and represented in Figure 5.5.

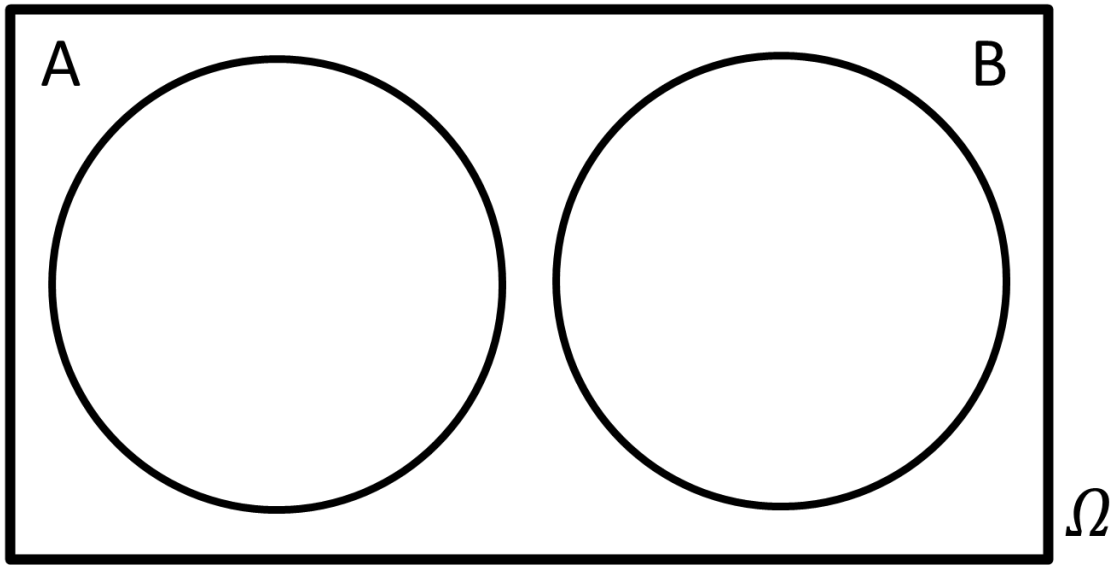


Figure 5.5. A representation of mutually exclusive events.

5.4.4 Probability of events

$\mathbb{DF}$: The probability of event A is the sum of probabilities of all sample points in A.

In situation where each element of the same sample space is equally likely to occur, we define the probability of an event A, $P(A)$ for short, in the following way:

$$P(A) = \frac{\text{Number of elements in the sample space that favour } A}{\text{Total number of elements in the sample space}} = \frac{N(A)}{k}$$

where k denotes the number of elements in the sample space.

Example 5.2. Consider the example of a well shuffled pack of cards. What is the probability that if we cut the pack, the card we obtain is a red king?

Solution

First of all, we need to identify the sample space. There are 52 elements in the sample space corresponding to 52 cards in the pack. How many of these elements favour getting a red king? Two – the King of Diamonds and the King of Hearts. Therefore, the probability of obtaining a red king is:

$$P(Red\ King) = \frac{2}{52} = 0.0385$$

5.4.5 Elementary properties of probabilities

In what follows let A denote any event, then:

i. $0 \leq P(A) \leq 1$, i.e. the probability of event lies in the range [0,1].
ii. If Ω represents the total sample space for a particular situation then $P(\Omega) = 1$. This is to say that one of the elements of the sample space must

occur- of this we are certain. An implication of this property in conjunction with property iii. below is that $P(A) + P(\bar{A}) = 1$.

iii. If two events, A and B are mutually exclusive then $A \cup B = P(A) + P(B)$. This is the addition law for mutually exclusive events.

5.4.6 The Addition Law for non-mutually exclusive events

$\mathbb{TH}$: If A and B are two events then $A \cup B = P(A) + P(B) - A \cap B$.

Proof

From the diagram in Figure 5.6 it is clear $A \cup B = A \cup (\bar{A} \cap B)$ and that two events A and $(\bar{A} \cup B)$ are mutually exclusive.

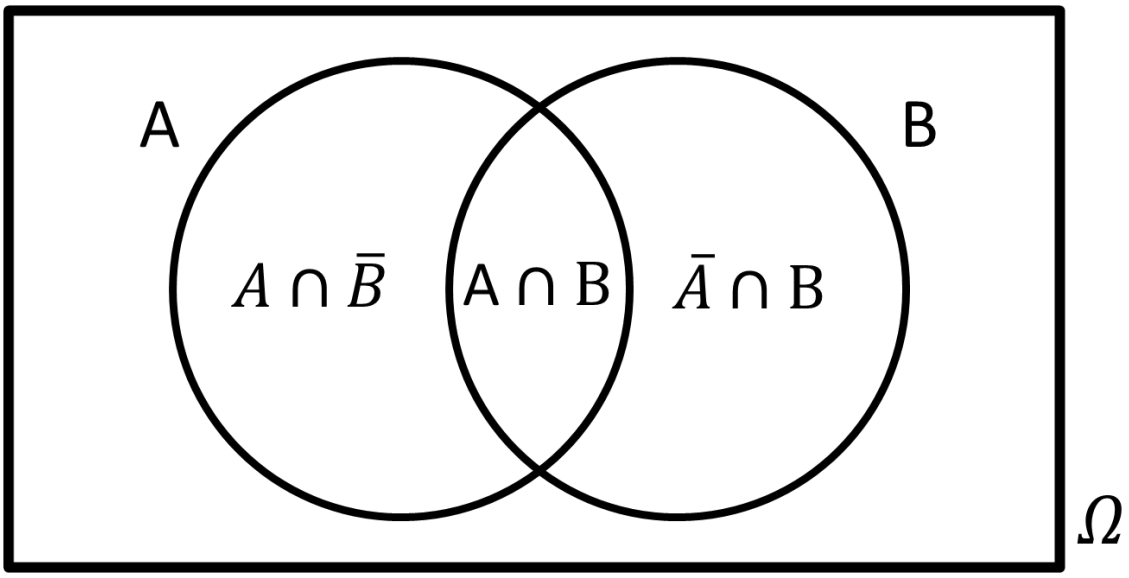


Figure 5.6. A Venn diagram representing Events A and B.

Therefore, using property iii. from above,

$$P(A \cup B) = P\big(A \cup (\bar{A} \cap B)\big) = P(A) + P(\bar{A} \cap B) \qquad (*)$$

From the diagram in Figure 5.6 it is also clear that $B = (A \cap B) \cup (\bar{A} \cap B)$ and again the two events $A \cap B$ and $\bar{A} \cap B$ are mutually exclusive. Therefore, again using property iii.,

$$P(B) = P\big((A \cap B) \cup (\bar{A} \cap B)\big) = P(A \cap B) + P(\bar{A} \cap B)$$

hence

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Substituting this results in (*) above we obtain:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

as required.

5.4.7 Conditional Probability

There are many situations that we can think of in which the occurrence of one event influences the occurrence of subsequent events.

$\mathbb{DF}$: The probability of an event B occurring when it is known that some event A has occurred is called a conditional probability and the following notation denotes the probability of B conditional upon the prior occurrence of A, or 'B given A' for short, $P(B|A)$. Now,

$$P(B|A) = \frac{\text{Number of events which favour the occurance of both } A \text{ and } B}{\text{Number of events which favour } A} = \frac{N(A \cap B)}{N(A)}$$

i.e. the sample space has been reduced due to the prior occurrence of A. But

$$\frac{N(A \cap B)}{N(A)} = \frac{N(A \cap B)/k}{N(A)/k} = \frac{P(A \cap B)}{P(A)}$$

where k denotes the number of elements in the sample space. Thus,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Similarly,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

5.4.8 Multiplication Law

The Multiplication Law for non-independent events follows directly from the above and is stated formally as:

$$P(A \cap B) = P(A)P(B|A)$$

or

$$P(A \cap B) = P(B)P(A|B)$$

providing that $P(A) > 0$. The Multiplication Law for independent events is just a special case of the above, but before we state it explicitly, we require a definition of independence.

Two events A and B are said to be independent if the occurrence, or non-occurrence, of A in no way influences the occurrence, or non-occurrence, B; and *vice versa*.

Formally, this means that if A and B are independent then:

$$P(B|A) = P(B)$$

and

$$P(A|B) = P(A)$$

Thus, for independent events, the Multiplication Law given above reduces to:

$$P(A \cap B) = P(A)P(B)$$

5.4.9 Law of total probabilities

$\mathbb{DF}$: Assume that $A_1, A_2, \ldots, A_n$ are mutually exclusive events ($A_i \cap A_j = \emptyset$ for $i \neq j$). Then for any event B its probability is defined as follows:

$$P(B) = \sum_{i=1}^{n} P(A_i)P(B|A_i)$$

which is called the Law of Total Probabilities.

5.4.10 Bayes' Theorem

We can amalgamate the Laws that have just been introduced to form the important Bayes' Theorem.

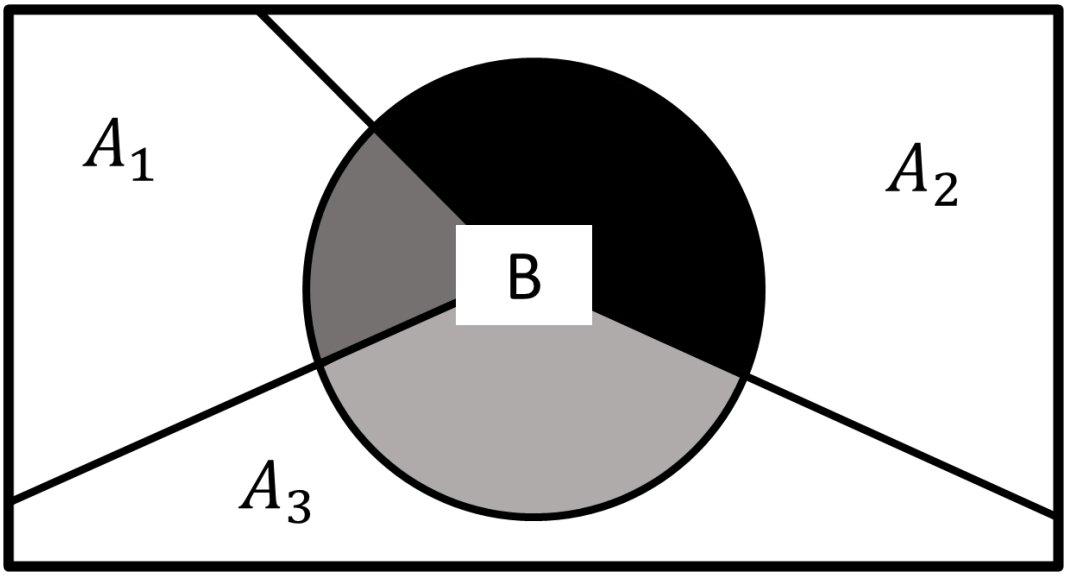


Figure 5.7. An illustration of Bayes' Theorem.

Derivation

Suppose we have three mutually exclusive events A_1, A_2 and A_3 (see diagram). Then we see that:

$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)$$

As these three contributing sections are mutually exclusive, it follows that:

$$P(B) = P[(A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)] = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)$$

Now using the Multiplication Law and the Law of Total Probabilities we have:

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

But,

$$P(A_1)P(B|A_1) = P(B)P(A_1|B)$$

and similar relations hold for A_2 and A_3. Therefore, the Bayes' Theorem states that:

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(B)} = \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)}$$

This result can be generalized to deal with any number of mutually exclusive events.

Example 5.3. A bag contains 10 beads; 5 white, 3 red and 2 green. Two beads are drawn, without replacement, from the bag. Find the probability that:

i. 2 whites,
ii. 2 of the same colour,
iii. 2 different colours are drawn.

Solution to i.

$$P(2 \text{ whites}) = P(\text{first bead is white and second is white})$$
$$= P(\text{first bead is white}) \times P(\text{second bead is white}|\text{first bead is white})$$

Note: the second line is a consequence of the Multiplication Law for non-independent events).

Now, $P(\text{first bead is white}) = 5/10$ and $P(\text{second is white}|\text{first is white}) = 4/9$. Therefore, $P(2 \text{ whites}) = 5/10 \times 4/9 = 20/90 = 2/9$

Alternative solution to i.

$$P(2 \text{ whites}) = \frac{\text{number of selections of 2 whites from 5 whites}}{\text{number of selections of 2 beads from 10}} = \frac{\binom{5}{2}}{\binom{10}{2}} = \frac{{}^5C_2}{{}^{10}C_2} = \frac{2}{9}$$

Solution to ii.

The possibilities in this case are, in an obvious notation, WW, RR and GG.

Therefore, $P(2 \text{ the same}) = P(WW \text{ or } RR \text{ or } GG) = P(WW) + P(RR) + P(GG)$. The latter is a consequence of the Addition Law for mutually exclusive events. Proceeding as in i.,

$$P(WW) = (5/10) \times (4/9)$$

$$P(RR) = (3/10) \times (2/9)$$

$$P(GG) = (2/10) \times (1/9)$$

Therefore,

$$P(2 \text{ the same}) = (5/10) \times (4/9) + (3/10) \times (2/9) + (2/10) \times (1/9) = 14/45$$

Solution to iii.

The simplest approach in this case is to realise that the event 'Draw two different colours' is the complement of the event 'Draw two beads of the same colour'. These

two events describe a complete sample space. Therefore, applying property ii. from the above,

$$P(\text{2 different}) + P(\text{2 the same}) = 1$$

Thus,

$$P(\text{2 different}) = 1 - P(\text{2 the same}) = 1 - (14/45) = 31/45$$

Example 5.4. Three factories X, Y and Z produce steel beams. From the past records it is found that 5% of the output from X is substandard, 3% from Y, and 2% from Z. The beams are stockpiled in a warehouse; 40% come from X, 50% come from Y and 10% come from Z. A beam is chosen at random and is found to be substandard. What is the probability that it came from factory X?

Solution

This is an example requiring the use of Bayes' Theorem. Using the notation of the derivation of the Theorem, let B represent the event 'beam is substandard'. Also, let A_1 represent the event 'beam comes from X', A_2 represent the event 'beam comes from Y' and A_3 represent the event 'beam comes from Z'.

We require $P(A_1|B)$ which, according to Bayes' Theorem, is given by:

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{\sum_{i=1}^{3} P(A_i)P(B|A_i)} = \frac{0.4 \times 0.05}{0.4 \times 0.05 + 0.5 \times 0.03 + 0.1 \times 0.02}$$
$$= \frac{0.02}{0.02 + 0.015 + 0.002} = 0.54$$

6 Random Variables, Expected Values and Probability Distributions

6.1 Learning Outcomes

At the end of this section, you must be able to:

- Define random variables as functions that assign numerical values to outcomes in a probability space.
- Differentiate between discrete and continuous random variables, grasping the distinctions and characteristics of each.
- Define and compute the expected value (mean) of a random variable.
- Define probability distributions as descriptions of the likelihood of different values for a random variable.

6.2 Introduction

Random variables serve as functionally crucial entities, mapping possible outcomes to numerical values in the realm of uncertainty. Digging deeper, the expected value becomes a pivotal measure, representing the average of potential outcomes and offering a systematic approach to navigate complex scenarios through the linearity of expectation. Meanwhile, probability distributions take center stage, presenting the mathematical structures that quantify the likelihood of different outcomes for a given random variable. From the discreet realm of binomial distributions to the continuous elegance of normal distributions, these tools lay the groundwork for analyzing uncertainty and making informed decisions. In this exploration, we delve into the technical intricacies of these concepts, aiming to equip ourselves with the mathematical precision required to unravel uncertainty and make strategic choices.

6.3 Basic Notation and Principles

Some definitions as follows:

$\mathbb{DF}$: A sample space (Ω) is the set of all possible outcomes of an experiment.

$\mathbb{DF}$: An event is a subset of the sample space.

$\mathbb{DF}$: A random variable (RV) is a function defined over the sample space Ω. It assigns a real value $X(e) = x$ to each possible outcome (event) e in Ω.

Note: Capital letters refer to random variables and lower case letters to the particular value that a random variable can take.

Example 6.1. Consider throwing two tetrahedral dice (or two throws of a single tetrahedral die) and consider the following random variables all involving the face on which the dice land. The various outcomes are:

Value on first face, value on second face		Value on second face			
		1	2	3	4
Value on first face	1	(1,1)	(1,2)	(1,3)	(1,4)
	2	(2,1)	(2,2)	(2,3)	(2,4)
	3	(3,1)	(3,2)	(3,3)	(3,4)
	4	(4,1)	(4,2)	(4,3)	(4,4)

We now consider various events.

V: the sum of two outcomes. V takes the values 2, 3, 4, 5, 6, 7, 8, i.e. $V = \{v: v \in 2, 3, 4, 5, 6, 7, 8\}$

W: higher score of the two dice $W = \{w: w \in 1, 2, 3, 4\}$

X: score of the first die $X = \{x: x \in 1, 2, 3, 4\}$

Y: the difference of scores (absolute values) $Y = \{y: y \in 0, 1, 2, 3\}$

Z: score on first die – score on the second die $Z = \{z: z \in -3, -2, \dots, 2, 3\}$

Each of these outcomes is an event and can be identified as an upper case letter or as a suffixed letter. We could define A_1 to be $v = 2$, i.e. the event A_1 is the event that the random variable V takes the value 2, and A_2 to be $v = 3$, etc. We could also have defined an event $A_2 = \{v \leq 3\}$, i.e. the event A_3 occurs if the random variable V takes the values 2 and 3.

6.4 Discrete Random Variables

$\mathbb{DF}$: If, as in the example above, the possible values are countable or infinitely countable, i.e. if $X = \{x \in x_1, x_2, \dots, x_n\}$ or $X = \{x \in x_1, x_2, \dots\}$ then X is said to be a discrete random variable.

For example, the number of items bought in a shop in a day, or the average number of tasks performed in a given period of time, are both examples of discrete random variables. However, if the number of outcomes is not at most countable infinity then the random variable is said to be continuous, e.g. heights of people.

6.4.1 Probability Distribution Function (Discrete Case)

A discrete random variable assumes each of its values with a certain probability.

$\mathbb{DF}$: A function $f(x) = P(X = x)$ for $x = x_1, x_2, \dots$ that assigns the probability to each possible value of x is called a discrete probability distribution function (p.d.f or pdf) or discrete probability mass function.

To be valid p.d.f f(x) must satisfy the following three properties:

i. $\forall x_i \; f(x_i) \geq 0$

ii. $\sum_{all\ x_i} f(x_i) = 1$

iii. $P(X = x_i) = f(x_i)$

We may express $f(x)$ either as a series of values (say, in a table) or by a formula.

Example 6.2. We return to Example 6.1 and consider the random variable V

v	2	3	4	5	6	7	8
Number of cases	1	2	3	4	3	2	1
	1/16	2/16	3/16	4/16	3/16	2/16	1/16

Alternatively, we can write:

$$f(v) = \begin{cases} \dfrac{v-1}{16} & v = 2,3,4,5 \\ \dfrac{9-v}{16} & v = (5),6,7,8 \\ 0 & \text{otherwise} \end{cases}$$

or

$$f(v) = \begin{cases} \dfrac{4-|v-5|}{16} & v = 2,3,4,5,6,7,8 \\ 0 & \text{otherwise} \end{cases}$$

This p.d.f can be also represented as a graph.

6.4.2 Cumulative Distribution Function (Discrete Case)

An alternative way of describing a random variable, X, is to specify its associated distribution function (or cumulative distribution function) which is usually denoted by $F(x)$. This function is defined as $F(x) = P(X \leq x)$, i.e. the probability of the random variable X being less than or equal to the value x. This is a very convenient way of describing a random variable since it can be used for both discrete and continuous distributions.

$\mathbb{DF}$: Distribution (cumulative distribution) function (c.d.f or c.d.f) is defined as $F(x) = P(X \leq x)$.

Note: It is customary to write the p.d.f as lower case ($f(x)$) and the c.d.f as upper case (($F(x)$).

The c.d.f is a monotone increasing function, i.e. one that never decreases as X increases. However, for a discrete p.d.f it is a stepped function, the size of the step at x is equal to $f(x)$ and we can see that:

$$F(x) = \sum_{X<x} f(x_i)$$

and that

$$f(x_i) = F(x_i) - F(x_{i-1})$$

Assuming that the x_i's are in ascending order.

Properties of c.d.f

i. $\lim_{x \to -\infty} F(x) = 0$

ii. $\lim_{x \to +\infty} F(x) = 1$

iii. $\lim_{h \to 0^+} F(x + h) = F(x)$

iv. $a < b \Rightarrow F(a) \leq F(b)$

v. $P(a < x \leq b) = F(b) - F(a)$

From the first, second and the fourth properties we infer that $0 \leq F(x) \leq 1$.

Example 6.3. If we have the following discrete distribution

x	0	1	2	3	4	5
$P(X = x)$	0.1	0.2	0.1	0.1	0.2	0.3

then we can compute the following cumulative properties in order to obtain the distribution function

x	0	1	2	3	4	5
$F(x) = P(x \leq x)$	0.1	0.3	0.4	0.5	0.7	1.0

And the cumulative distribution function F(x), is a step-function as represented by the diagram in Figure 6.1.

Do it in **R**:

```
#x values
x <- 0:6
#probabilities
fx <- c(0, 0.1, 0.2, 0.1, 0.1, 0.2, 0.3)

Fx <- cumsum(fx)
n <- length(x)

#creare empty plot
plot(x = NA, y = NA, pch = NA,
     xlim = c(0, 5.7),
     ylim = c(0, 1),
     xlab = "x",
     ylab = "F(x)",
     main = "",
```

```
     bty="n")

points(x = x[-n], y = Fx[-1], pch=19)
points(x = x[-1], y = Fx[-1], pch=1)
for(i in 1:(n-1)) points(x=x[i+0:1], y=Fx[c(i,i)+1], type="l")
```

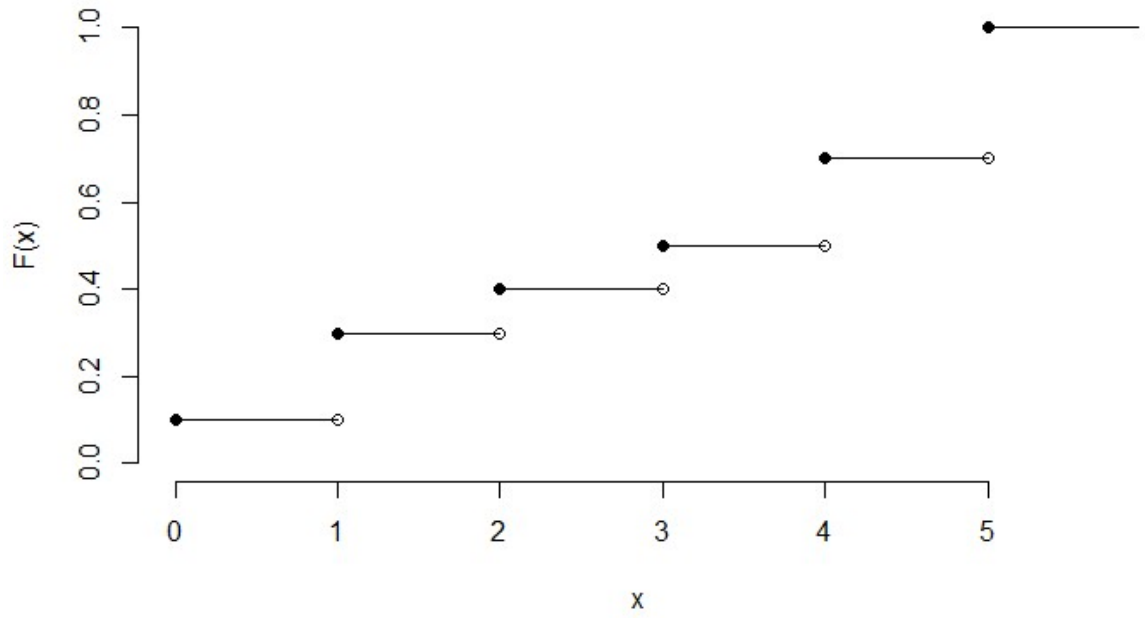


Figure 6.1. The c.d.f that was given in Example 6.3.

6.4.3 Mean or expected value (Discrete Case)

$\mathbb{DF}$: If X is a discrete random variable with p.d.f then the *expected value* (or mean) is defined by:

$$\mu = E[X] = \sum_{all\ x_i} x_i \times f(x_i)\text{, if it exist.}$$

So if it is an infinite series and is not absolutely convergent then we say that $E[X]$ does not exist. We write $E[X]$ variously as $\mu, \mu'$and μ_1'; the particular one used will often be determined by the circumstances. We shall look at a wider application of this concept of expectation later.

Example 6.4. Consider the distribution of the higher score of the two tetrahedral dice.

What is the mean of this distribution?

x	1	2	3	4
$f(x)$	1/16	3/16	5/16	7/16

Solution

The mean is:

$$E[X] = \sum_{all\ x_i} x_i \times f(x_i) = 1 \times \frac{1}{16} + 2 \times \frac{3}{16} + 3 \times \frac{5}{16} + 4 \times \frac{7}{16} = \frac{1+6+15+28}{16} = \frac{50}{16}$$

$$= 3\frac{1}{8}$$

6.5 Continuous Random Variables

A continuous random variable has an infinite number of values and it has a probability of 0 of assuming exactly any of its values. Consequently, its probability distribution cannot be given in tabular form.

Let us consider a random variable whose values are the heights of all people over 21 years of age. Between any two values, say 164.5 cm and 165.5 cm there are an infinite number of heights, one of which is 165 cm. The probability of selecting a person with height exactly 165 cm but not one of the infinitely large set of heights so close to 165 is humanly impossible and therefore we assign a probability 0 to the event “164 cm high”.

We shall deal with computing probabilities for various intervals of continuous random variables such as $P(a < X < b) = P(a < X < b) + P(X = b) = P(a < X < b)$ so it does not matter whether the end points of the interval included or not (Note that this is not true for discrete random variable).

6.5.1 Probability Distribution Function (The Continuous Case)

The probability density function $f\ (x)$ for continuous random variable X is constructed in a way that the area under the curve is equal to 1 when computed over the range of X for which $f\ (x)$ is defined.

From calculus we know that:

$$P(a < X < b) = \int_a^b f(x)dx$$

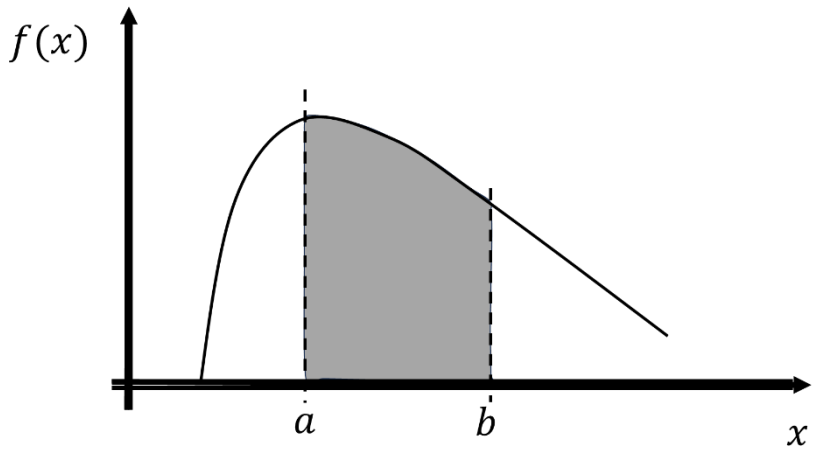


Figure 6.2. An illustration of integration with boundaries to calculate the area.

$\mathbb{DF}$: The function $f(x)$ is a probability density function for the continuous random variable X, defined over the set of real numbers R, if:

i. $\forall x \in R\ f(x) \geq 0$, i.e. the function is non-negative.

ii. $\int_{-\infty}^{+\infty} f(x)dx = 1$, i.e. the total area under the p.d.f is equal to 1.

iii. If a and b are two values and $a \leq b$ then $P(a < X < b) = \int_a^b f(x)dx$, i.e. the probability of obtaining a value which lies between limits a and b is given by the integral of the p.d.f with limits a and b.

6.5.2 Cumulative Distribution Function (Continuous Case)

$\mathbb{DF}$: The [cumulative] distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is:

$$F(x) = \int_{-\infty}^{x} f(t)dt \text{ for } -\infty < x < +\infty$$

Note 1. $F(x)$ and $f(x)$ may be piecewise.

Note 2. It follows from the fundamental theorem of calculus that $f(x) = \frac{dF(x)}{dx}$ if it exits.

The implication of this definition is that $P(a < X < b) = F(b) - F(a)$.

Properties of cumulative distribution functions

For any cumulative distribution function, $F(x)$ say, the following two properties hold:

i. $F(x)$ is non-negative for any value of x, i.e. $F(x) \geq 0$ for all x.

ii. $F(-\infty) = 0$ and $F(+\infty) = 1$, i.e. a distribution function has minimum value 0 and maximum value 1.

Example 6.5. Find the value of the constant c and $F(x)$ for the distribution:

$$f(x) = \begin{cases} ce^{-x} & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Solution

$$F(x) = \int_0^x f(t)dt = \frac{ce^{-t}}{-1}\Big|_0^x = \frac{ce^{-x}}{-1} - \frac{ce^0}{-1} = c(1 - e^{-x})$$

We know that $F(1) = 1$.

$$\therefore c(1 - e^{-1}) = 1$$

$$\therefore c = \frac{1}{1 - e^{-1}}$$

So, we now have:

$$f(x) = \begin{cases} \dfrac{e^{-x}}{1-e^{-1}} & 0 \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

and

$$F(x) = \begin{cases} 0 & x < 0 \\ \dfrac{1-e^{-x}}{1-e^{-1}} & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

Note 1: If we were not required to find the c.d.f then we would have integrated over the full range of the distribution and equated this to 1.

Note 2: If the distribution is piecewise then deal with each piece individually.

6.5.3 Mean or Expected value (Continuous Case)

$\mathbb{DF}$: If X is a random variable with p.d.f $f(x)$ then:

$$\mu = E[X] = \begin{cases} \sum_{all\ x} xf(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} xf(x), & \text{if } X \text{ is continuous} \end{cases}$$

Note that this interchange of summation for discrete and integration for continuous distributions is quite general.

$\mathbb{DF}$: If X is a random variable with p.d.f $f(x)$ then the expected value of the random variable $u(X)$ is

$$E[u(X)] = \begin{cases} \sum u(x) \times f(x) & \text{if } X \text{ is a discrete random variable} \\ \int u(x) \times f(x) & \text{if } X \text{ is a continuous random variable} \end{cases}$$

Properties of expected values

$$E[u(X) + v(X)] = E[u(X)] + E[v(X)]$$

since

$$\int \{u(x) + v(x)\} f(x) dx = \int u(x) f(x) dx + \int v(x) f(x) dx$$

More generally,

$$E[a \times u(X) + b \times v(X)] = a \times E[u(X)] + b \times E[v(x)]$$

And specifically,

$$E[aX + b] = aE[X] + b$$

The most important expected value is that of the random variable X itself. This is called the mean. The second important measure is the variance. i.e. Var(X) or σ^2

$\mathbb{DF}$: Let X be a random variable with p.d.f $f(x)$ and mean μ. The variance of X is:

$$\sigma^2 = \text{Var}(X) = E[(X-\mu)^2] = \begin{cases} \sum_{\text{all } x} (x-\mu)^2 f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x-\mu)^2 f(x) & \text{if } X \text{ is continuous} \end{cases}$$

$$\text{Var}(X) = E[(X-\mu)^2] = E[X^2 - 2X\mu + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$
$$= E[X^2] - 2\mu\mu + \mu^2 = E[X^2] - \mu^2$$

Thus,

$$\text{Var}(X) = E[X^2] - \mu^2 = E[X^2] - \{E[X]\}^2$$

The latter form is often used since it is numerically easier to compute. The variance of X is used as a measure of variability. If $Var(X) > Var(Y)$ then X is more variable than Y in the values that it takes. Conversely, the Y values are more tightly clustered around their mean.

Note that, $\text{Var}(aX+b) = a^2\text{Var}(X)$

Proof

$$\text{Var}(aX+b) = E[\{aX+b-E[aX+b]\}^2] = E[\{aX+b-aE[X]-b\}^2] = E[a^2(X-\mu)^2]$$
$$= a^2 E[(X-\mu)^2] = a^2\text{Var}(X)$$

Standard Deviation

If X has units such as kilos then $\text{Var}(X)$ has units kilo^2. A convenient way round this difficulty is the define standard deviation such as

$\mathbb{DF}$: $SD = STD = ST = \sigma = +\sqrt{\text{Var}(X)}$

Example 6.6. Consider a truncated exponential distribution

$$f(x) = \begin{cases} \dfrac{e^{-x}}{1-e^{-1}} & 0 \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

We calculate:

$$E[X] = \int_0^1 x f(x) dx$$

and

$$E[X^2] = \int_0^1 x^2 f(x) dx$$

which, after some calculus, can be shown to be 0.4180 and 0.2540 respectively. Thus, the mean is 0.4189 and the variance is $0.2540 - (0.41802)^2 = 0.0793$ (and therefore the standard deviation is 0.2816).

6.5.4 The median and the mode of a continuous distribution

Suppose we consider a continuous random variable X with p.d.f $f(x)$. The mode of the distribution is defined to be that value of x for which $f(x)$ is a maximum. The median of the distribution is defined to be a value, M say, which divides the area under the p.d.f in half, i.e.

Area under the p.d.f to the left of the median=Area under the p.d.f to the right of the median $= 1/2$

or

$$P(X < \text{Median}) = P(X > \text{Median}) = \frac{1}{2}$$

or

$$\int_{-\infty}^{M} f(x)dx = \frac{1}{2} = \int_{M}^{\infty} f(x)dx$$

This implies that in the diagram below $A| = |B| = 1/2$, where $|A|$ denotes the area of the region A.

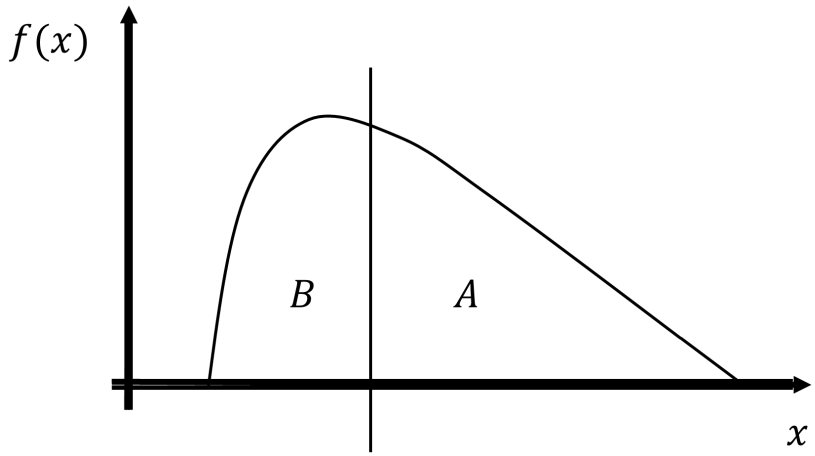


Figure 6.3. An illustration of obtaining the median of a p.d.f

Example 6.7. Given

$$f(x) = \begin{cases} kx(1-x) & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

i. Find k.

ii. Calculate the mean, variance and standard deviation.

iii. Calculate the median and mode.

Solution to part i.

We know that the integral of the p.d.f is 1 for any valid p.d.f and hence

$$\int_{-\infty}^{+\infty} f(x)dx = \int_0^1 kx(1-x)dx = k\left\{\int_0^1 kx(1-x)dx\right\} = 1$$

Thus,

$$k\frac{x^2}{2} - \frac{x^3}{3}\Big|_0^1 = 1$$

$$\frac{k}{6} = 1$$

Hence, $k = 6$

Solution to part ii.

Substituting k into initial formula for the p.d.f, the mean of the distribution is given by:

$$\mu = E[X] = \int_{-\infty}^{+\infty} xf(x)dx = \int_0^1 x6x(1-x)dx = 6\frac{x^3}{3} - \frac{x^4}{4}\Big|_0^1 = \frac{6}{12} = \frac{1}{2}$$

The variance of the distribution is given by:

$$\sigma^2 = Var(X) = E[X^2] - \{E[X]\}^2 = \int_0^1 x^2 6x(1-x)dx - \left\{\frac{1}{2}\right\}^2 = 6\frac{x^4}{4} - \frac{x^5}{5}\Big|_0^1 - \frac{1}{4}$$
$$= \frac{6}{20} - \frac{1}{4} = \frac{1}{20}$$

The standard deviation is therefore given by $\sqrt{\frac{1}{20}}$

Solution to part iii.

The median is that value, M say, for which

$$\int_0^M f(x)dx = \int_0^M 6x(1-x)dx = \frac{1}{2}$$

Therefore,

$$6\left\{\frac{x^2}{2} - \frac{x^3}{3}\right\}\Big|_0^M = \frac{1}{2}$$

i.e.

$$6\left\{\frac{M^2}{2} - \frac{M^3}{3}\right\} = 3M^2 - 2M^3 = M^2[3-2M] = \frac{1}{2}$$

The solution of to this equation is $M = 1/2$, i.e. the median is ½.

Finally, the mode of the distribution is obtained by finding the value of x for which the p.d.f is a maximum. In this case, because the function is well behaved, this can be done by differentiating the p.d.f with respect to x and finding this value of x for which the derivative is 0 – in order to find the value of x for which the p.d.f is a maximum. Of course, we also have to check that the value obtained really is a maximum and not a minimum or point of inflexion.

In this case,

$$\frac{df(x)}{dx} = \frac{d\{6x(1-x)\}}{dx} = \frac{d\{6x-6x^2\}}{dx} = 6 - 12x$$

Setting the derivative equal to zero, we obtain $6 - 12x = 0$, i.e. $6 = 12x$ and hence $x = 1/2$.

To check that this is truly a maximum we note that the second derivative of the p.d.f is -12 which confirms our value of $x = 1/2$ as being the maximum value of the p.d.f Thus the mode of the distribution is the value of $x = 1/2$. Note that the method of identifying the mode of a distribution by finding the turning point of the p.d.f cannot be used generally. In practice, make sure you sketch the p.d.f or have some idea of what the p.d.f looks like before you leap in and attempt to use the above method for finding maxima!

6.6 Other Measures of Variability

Higher Moments

$\mathbb{DF}$: The kth moment about the origin is denoted by μ'_k and is defined $\mu'_k = E[X^k]$ whilst the kth moment about the mean, denoted by μ_k, is defined as $\mu_k = E[(X-\mu)^k]$ where μ is the mean. Indeed, we can see that the mean could be written as μ'_1 and the variance as μ_2. It is obvious that μ_1 always takes the value 0.

If the distribution is symmetrical about c, i.e. $f(x-c) = f(x+c)$ then $\mu'_1 = c$, and all odd moments about the mean are zero. Because of this we can use μ_3 as a measure of skewness of a distribution.

However, it should be noted that whilst the third central moment of a symmetrical distribution is zero, the converse is not true. It is possible to construct a distribution which is not symmetrical and its $\mu_3 = 0$. Nonetheless, we use μ_3 as a measure of skewness. Because of dimensionality of this we modify it to make it dimensionless, i.e. we make it into a coefficient.

7 Discrete Probability Distributions

7.1 Learning Outcomes

At the end of this section, you must be able to:

- To grasp the fundamental concepts of discrete probability distributions, including the definition of random variables, probability mass functions (PMFs), and cumulative distribution functions (CDFs).
- To apply probability formulas to solve problems involving discrete probability distributions, such as calculating expected values, variances, and standard deviations.
- To interpret the results of probability distributions in real-world contexts

7.2 Introduction

Often the observations generated by different statistical experiments have the same general type of behavior. In fact, one needs just a handful of important probability distributions to describe many of the discrete random variables encountered in practice. We are going to study some of them.

7.3 Discrete Uniform Distribution

$\mathbb{DF}$: If the random variable X assumes the values $x_1, x_2, \ldots, x_N$ with equal probabilities, then the discrete uniform distribution is given by:

$$f(x) = \frac{1}{N}$$

where $x = x_1, x_2, \ldots, x_N$. Usually, the following notation is used to indicate that the uniform distribution depends on parameter N: $U(x; N)$.

The mean of $X \sim U(x: N)$ is:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

The variance of $X \sim U(x: N)$ is:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N} (x_i - \mu)^2$$

Example 7.1. A classic example of a discrete uniform distribution is rolling a fair six-sided die. In this case there are six possible outcomes: 1, 2, 3, 4, 5 and 6. Each outcome has a probability of 1/6 of occurring when the die is rolled. The probability of getting any particular number on the die is the same for all numbers. See below the probability distribution table for rolling a fair six-sided die:

Outcome	Probability

1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Table 7.1. All possible outcomes and corresponding probabilities of rolling a fair six-sided die.

The plot of this is given in Figure 7.1. The equal bar lengths shows that each outcome has equal probabilities.

Do it in **R**:

```
# Define the outcomes and their probabilities
outcomes <- 1:6
probabilities <- rep(1/6, 6)

# Create a bar plot
barplot(probabilities, names.arg = outcomes, col = "skyblue",
        main = "Discrete Uniform Distribution of Rolling a Fair Six-sided
Die",
        xlab = "Die Face", ylab = "Probability", ylim = c(0, 0.20))
```

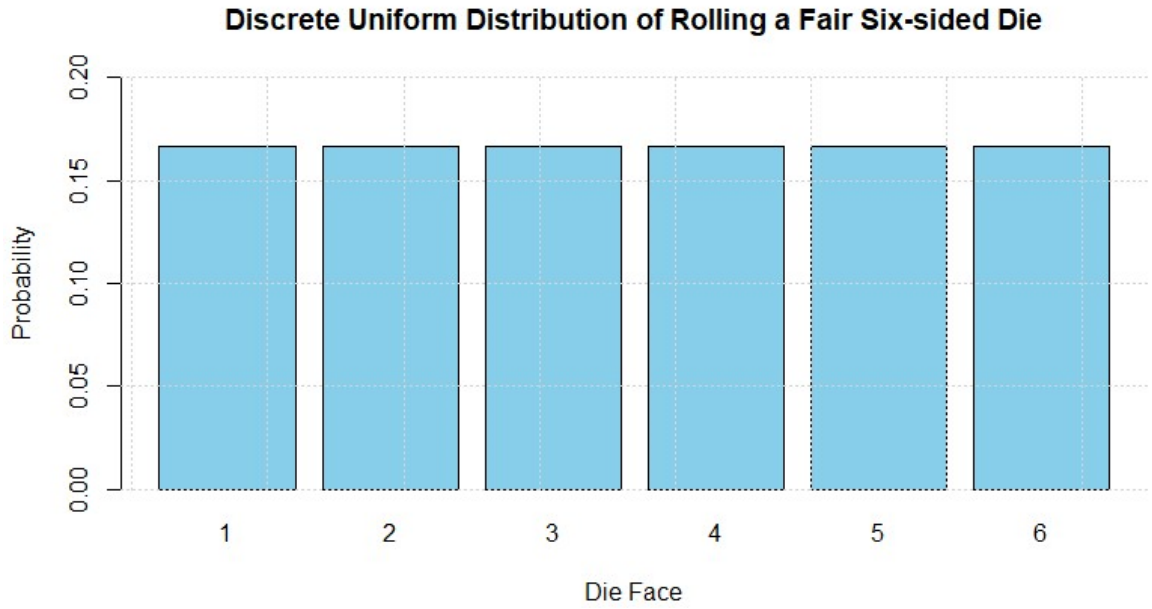


Figure 7.1. An illustration of a Uniform Distribution. This represents a fair 6-sided die with equal probability.

7.4 Binomial Distribution

An experiment often consists of repeated trials, each with two possible outcomes that may be labelled as “success” and “failure”. This process is referred to as Bernoulli process and each trial is called a Bernoulli trial.

Strictly speaking, the Bernoulli process must possess the following properties:

- the experiment consists of *n* repeated trials;
- each trial results in an outcome that may be classified as a success or a failure;
- the probability of success, *p*, remains constant from trial to trial;
- the repeated trials are independent.

$\mathbb{DF}$: If a Bernoulli trial can result in a success with probability p and a failure with probability $q = 1 - p$ then the p.d.f of the binomial random variable X, the number of successes in n independent trials, is:

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}$$

where $x = 0,1,2, \dots, n$

The mean of $X \sim b(x; n, p)$ is:

$$\mu = np$$

The variance of $X \sim b(x; n, p)$ is:

$$\sigma^2 = np(1-p) = npq$$

Example 7.2. The probability that a patient recover from a rare blood disease is 0.4. If 15 people are known to contract this disease, what is the probability that

i. at least 10 survive;

ii. from 3 to 8 survive;

iii. and exactly 5 survive?

Solution to part i.

$$P(X \geq 10) = P(X = 10) + P(X = 11) + \cdots + P(X = 15) = 0.0245 + 0.0074 + \cdots$$
$$= 0.0338$$

Do it in **R**:

```
# Probability of at least 10 surviving
prob_at_least_10 <- 1 - pbinom(9, size = 15, prob = 0.4)

# Display the probability
prob_at_least_10
```

```
## [1] 0.0338333
```

Solution to part ii.

$$P(3 \leq X \leq 8) = P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) + P(X = 8)$$
$$= 0.0634 + 0.1268 + \ldots + 0.1181 = 0.8779$$

Do it in **R**:

```
# Probability that between 3 and 8 survive
prob_3_to_8 <- sum(dbinom(3:8, size = 15, prob = 0.4))

# Display the probability
prob_3_to_8
```

```
## [1] 0.8778386
```

Solution to part iii.

$$P(X = 5) = 0.1859$$

Do it in **R**:

```
# Probability that exactly 5 survive
prob_exactly_5 <- dbinom(5, size = 15, prob = 0.4)

# Display the probability
prob_exactly_5
```

```
## [1] 0.1859378
```

For the probabilities obtained in part ii. and iii., they can be obtained using **R** or using the Binomial formula or from the table of Binomial probabilities. For the latter, maybe try it out a few times, but in recent years we tend to reply on computers. Since tables are pre-computed and rounded estimates, computers tend to be more accurate. This also applies to all the other probability distributions explored in this module.

Do it in **R**:

```
# Set the parameters
n <- 15  # Number of trials (people)
p <- 0.4  # Probability of survival for each person

# Create a vector of possible outcomes (number of survivors)
x <- 0:n

# Calculate the PMF for each possible outcome
pmf <- dbinom(x, n, p)

# Create a bar plot to illustrate the PMF
barplot(pmf, names.arg = x, main = "Binomial distribution for Rare Blood D
isease Survival",
        xlab = "Number of Survivors", ylab = "Probability", col = "lightbl
ue", border = "black", ylim = c(0, 0.25))
```

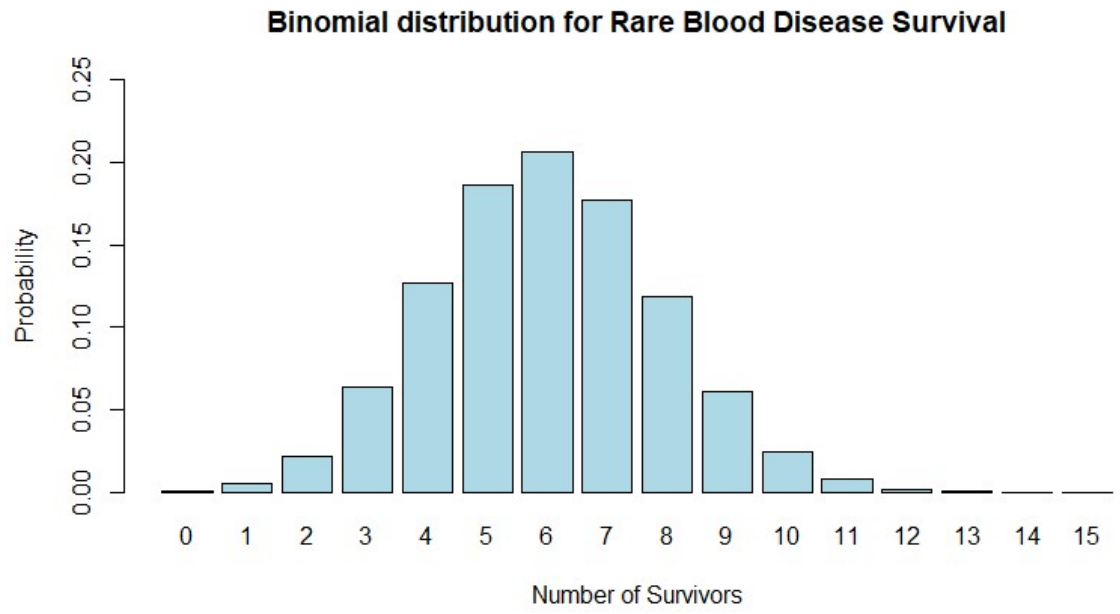


Figure 7.2. Binominal distribution corresponding to Example 7.2.

7.5 Negative Binomial Distribution

Let us consider an experiment where the properties are the same as before with the exception that the trials are repeated until the fixed number of successes occurs. Instead of finding the probability of x successes in n trials (when n is fixed), we are interested in the probability that rth success occurs on the xth trial. Experiments of this kind are called negative binomial experiments.

$\mathbb{DF}$: The number of X of trial to produce r success in a negative binomial experiment is called the negative binomial random variable.

Individual probabilities are obtained using the formula as follows:

$$P(X = x) = {}^{x-1}C_{r-1}p^r q^{x-r} \quad \text{where } x = r, r+1, \ldots.$$

To see how this formula is derived we note that to obtain rth success on the xth trial we must have obtained $(r-1)$ successes in the preceding $(x-1)$ trails. The final success occurs on the xth trial.

The probability of observing $(r-1)$ successes in $(x-1)$ independent trials is simply a binomial distribution and is given by

$$P\big((r-1) \text{ successes in } (x-1) \text{ trials}\big)$$
$$= {}^{x-1}C_{r-1}p^{r-1}(1-p)^{(x-1)-(r-1)} = {}^{x-1}C_{r-1}p^{r-1}(1-p)^{x-p} = {}^{x-1}C_{r-1}p^{r-1}q^{x-p}$$

We obtain a success on the final trial with probability p. The two events occur independently so the probability of the first event occurring and the second event occurring is simply a product of the two probabilities outlined above. Hence,

$$P(X = r) = (p)^{x-1}C_{r-1}p^{r-1}q^{x-r} = {}^{x-1}C_{r-1}p^r q^{x-r}$$

$\mathbb{DF}$: If repeated independent trials can result in a success with probability p and a failure with probability, $q = 1 - p$, then the probability distribution of the negative binomial random variable X is:

$$b^*(x; r, p) = NB(r, p) = \binom{x-1}{r-1} p^r q^{x-r} \quad \text{where } x = r, r+1, r+2, \ldots$$

Example 7.3. The probability that a person living in a certain city owns a dog is 0.3. Find the probability that the 10th person randomly interviewed in the city is the 5th one to own a dog.

$$b^*(10; 5, 0.3) = \binom{10-1}{5-1} 0.3^5 0.7^{10-5} = \frac{9!}{4!\,(9-4)!} 0.3^5 0.7^5 = 0.0515$$

The mean of $X \sim b^*(x; r, p)$ is $\mu = \frac{r}{p}$.

The variance of $X \sim b^*(x; r, p)$ is $\sigma^2 = \frac{r(1-p)}{p^2}$

Do it in **R**:

```
# Given values
p <- 0.3    # Probability of owning a dog
k <- 5      # 5th person to own a dog
n <- 10     # 10th person interviewed

# Calculate the probability using the negative binomial distribution
prob_10th_is_5th <- dnbinom(n - k, size = k, prob = p)

# Display the probability
prob_10th_is_5th
```

```
## [1] 0.05145967
```

The probability that the 10th person randomly interviewed in the city is the 5th one to own a dog is approximately 0.05145967, or 5.15%.

7.6 Geometric Distribution

The Geometric distribution is a special case of Negative Binomial distribution when $r = 1$.

Example 7.4. In an experiment we are tossing a coin until a head occurs. We are interested in the probability that the first head occurs on the 4th toss. Using Negative Binomial distribution, we have:

$$b^*(x; 1, p) = \binom{x-1}{1-1} p^1 q^{x-1} = pq^{x-1} \quad \text{where } x = 1, 2, 3, \ldots$$

Since the successive terms for $x = 1, 2, 3, \ldots$ constitute a geometric progression, it is usually called a geometric distribution denoted by $g(x; p)$. For example:

$$g(x; p) = g(4; 0.5) = b^*(4; 1, 0.5) = \binom{4-1}{1-1} 0.5^1 0.5^3 = 0.5 \times 0.5^3 = 0.0625$$

If the first success occurs at the xth trail, then it must have been preceded by $(x - 1)$ failures. Clearly, all such probabilities are non-negative, and they sum to 1 as it is shown below.

For $0 < p < 1$

$$\sum_{x=1}^{\infty} P(X = x) = \sum_{x=1}^{\infty} p(1-p)^{x-1}$$

$$= p \sum_{x-1}^{\infty} (1-p)^{x-1} = p(1 + q + q^2 + \cdots) = \frac{p}{1-q} = \frac{p}{p} = 1$$

Do it in **R**:

```r
# Given values
p <- 0.5   # Probability of getting a head
k <- 1     # We want the first success (k = 1)
n <- 4     # On the 4th toss

# Calculate the probability using the Negative Binomial distribution
prob_4th_is_1st_head <- dnbinom(n - k, size = k, prob = p)

# Display the probability
prob_4th_is_1st_head
```

```
## [1] 0.0625
```

7.7 Poisson Distribution

Experiments that produce numerical values of a random variable X, the number of outcomes occurring during a given time interval or in a specific region, are called Poisson experiments. Examples of Poisson experiments are:

- number of telephone call received by the office per hour;
- number of typing errors on page;
- number of bacteria in a given culture.

Properties of Poisson process

A Poisson experiment is derived from the Poisson process and possesses the

following properties:

- The number of outcomes in each time interval is unlimited.
- The number of outcomes that occur in one time interval is independent of the number of outcomes that occur in any other disjoint time intervals (we can say that Poisson process has no memory).
- Rate of events λ is constant meaning that on average each interval got λt events detected.
- The probability that more than one outcome will occur is a short time interval Δt is negligible.

$\mathbb{DF}$: The number X of outcomes occurring during a Poisson experiment is called a Poisson random variable and its probability distribution is called the Poisson distribution.

The number of outcomes in each time interval depends on rate of events λ, so the probability will depend on λ; we will denote it as $p(x;\lambda t)$.

$\mathbb{DF}$: The Poisson distribution is defined as:

$$p(x;\lambda t) = \frac{e^{-\lambda t}(\lambda t)^x}{x!}$$

where $x = 0,1,2,\ldots$ where λ is the average number of outcomes per unit time, e is a constant 2.71828…,

Both the mean and the variance of the Poisson distribution $X \sim p(x;\lambda t)$ are:

$$\mu = \sigma^2 = \lambda t$$

Example 7.5. The average number of customers buying coffee every 10 minutes is 4. What is the probability that 6 customers buy coffee in a given 10 minutes?

Let $x = 6$ and $\lambda t = 4$, thus:

$$p(6;4) = \frac{e^{-4}(4)^6}{6!} = 0.1042\ (to\ 4\ d.p)$$

We can extend this example by considering other values of x. Substituting other values of x, we obtain table 7.2.

Number of customers buying coffee in 10 minutes (x)	0	1	2	3	4	5	6
$p(x;\lambda t)$	0.0183	0.0733	0.1465	0.1954	0.1954	0.1563	0.1042

Table 7.2. Probabilities for a Poisson Distribution for $x = 0,1,2,3,4,5,6$

Do it in **R**:

```
# Given values
lambda <- 4  # Average number of customers every 10 minutes
k <- 6       # Number of customers we are interested in

# Calculate the probability using the Poisson distribution
probability_6_customers <- dpois(k, lambda)

# Display the probability
probability_6_customers
```

```
## [1] 0.1041956
```

Do it in **R**:

```
# Given values
lambda <- 4  # Average number of customers every 10 minutes

# Range of customer counts to consider (0 to 10)
customers <- 0:10

# Calculate the probability for each number of customers
probabilities <- dpois(customers, lambda)

# Plot the probabilities
plot(customers, probabilities, type = "h", lwd = 2, col = "blue",
     main = "Poisson Distribution of Customers Buying Coffee",
     xlab = "Number of Customers",
     ylab = "Probability",
     ylim = c(0, max(probabilities) * 1.1))
```

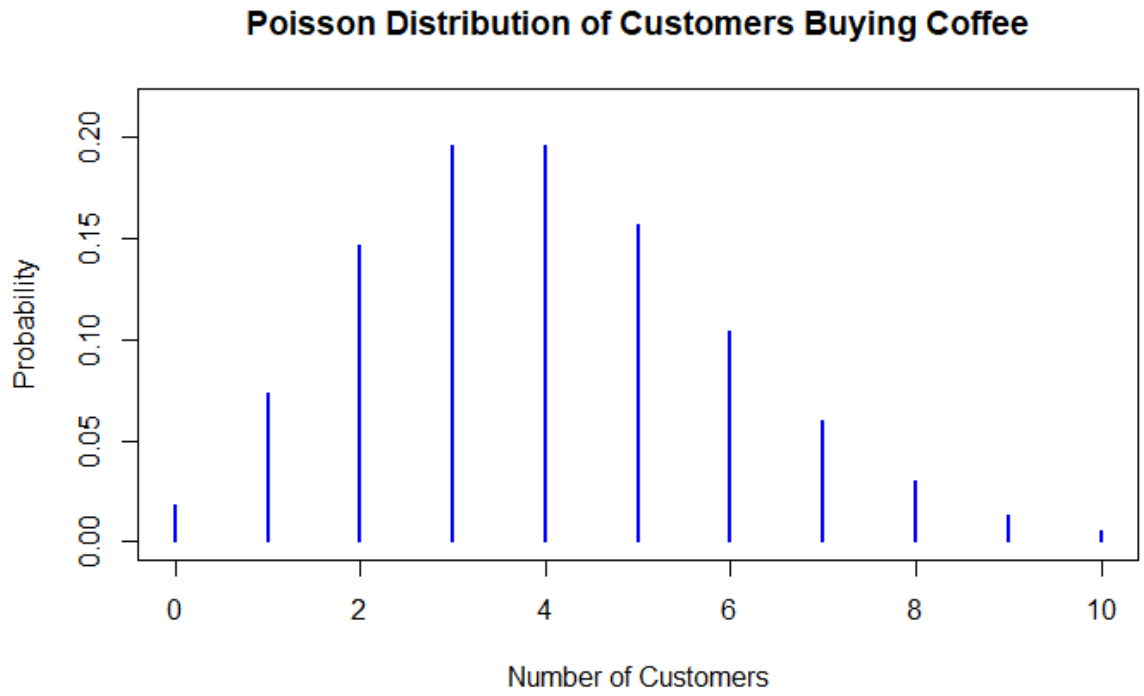


Figure 7.3. Graphical representation of example 7.5

8 Continuous Probability Distributions

8.1 Learning Outcomes

At the end of this section, you must be able to:

- Gain familiarity with common continuous probability distributions, such as the normal distribution, exponential distribution, uniform distribution, and others.
- Understand the characteristics, properties, and parameters associated with each distribution.
- Use probability density functions to calculate probabilities associated with specific intervals or events.

8.2 Introduction

In the last chapter we considered discrete distributions. The Binomial and Poisson distributions are very useful as they can be applied to many situations arising in practice which are themselves characterized by producing discrete data. In this lecture we consider those distributions which can be used to describe situations which produce continuous data.

As an example, consider a study into the wing-span measurements of long eared bats. Clearly the data arising out of such a study will be continuous in nature. If we wanted to describe the sample space, it would consists of all those numbers in an interval along the Real Line; perhaps [7, 17] cm. It would not consist of a collection of discrete elements with gaps between them as was the case for discrete distribution we considered earlier.

A consequence of this is that we have to abandon the notion of probability for each element of the sample space as was defined in the case of discrete distribution. In fact, the probability of obtaining any particular value from a continuous distribution is 0. Instead, a continuous distribution is defined by a function known as a probability density function (p.d.f). This function represents how likely (we use 'likely' here according to its standard usage in the English language – it most definitely should not be interpreted as meaning the same thing as the technical term 'probability') a particular value, say x, is to occur. Given such a function it is, however, possible to talk about the probability of obtaining a value between any two limits.

Recall that a p.d.f is usually denoted by a lower-case letter f and is expressed as a function of x, i.e. $f(x)$.

8.3 Continuous Uniform Distribution

Uniform distribution is characterised by probability density function which is "flat" and thus the probability is uniform in a closed interval $[a, b]$.

$\mathbb{DF}$: The probability and density functions of a continuous uniform random variable X

on the interval $[a, b]$ are

$$f(x) = \begin{cases} \dfrac{1}{b-a} & a < x < b \\ 0 & \text{elsewhere} \end{cases}$$

and

$$F(x) = \begin{matrix} 0 & \text{for } x < a \\ \dfrac{x-a}{b-a} & \text{for } a \le x \le b \\ 1 & \text{for } x > b \end{matrix}$$

The graphs of these are given below:

a) b)

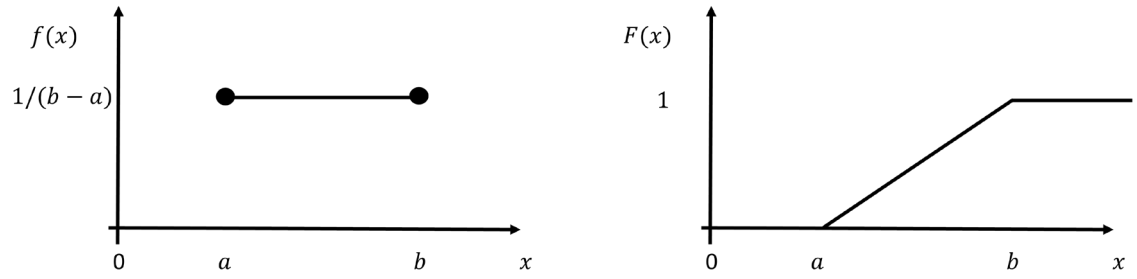


Figure 8.1. An illustration of a continuous uniform distribution.

This function is very simple to interpret given our definition of a p.d.f It simply implies that every element in the range (a, b) is as likely as any other and that any value outside that range has no chance of occurring.

If we consider the $U(a, b)$ as a probability density function, then it should satisfy the following conditions:

i. Clearly, $f(x) \geq 0$ for all x

ii. $\int_a^b \frac{1}{b-a} dx = \frac{x}{b-a}\Big|_a^b = \frac{b-a}{b-a} = 1$, i.e. $U(a, b)$ is a distribution function.

iii. For $P(x_1 < X < x_2) = \int_{x_1}^{x_2} f(x)dx = \int_{x_1}^{x_2} \frac{1}{b-a} dx = \frac{1}{b-a} x\Big|_{x_1}^{x_2} = \frac{x_2 - x_1}{b-a}$

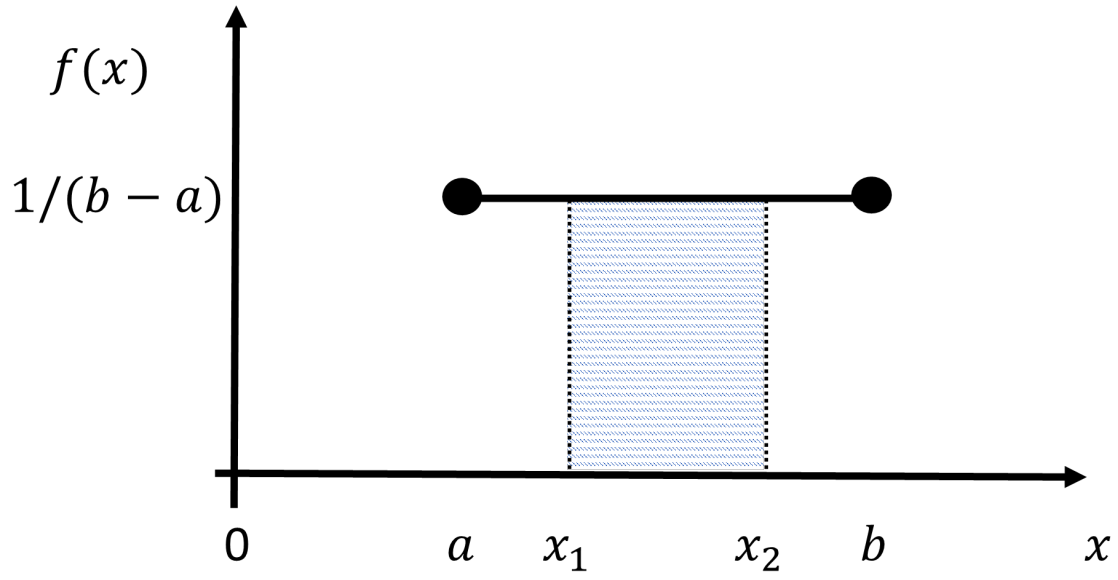


Figure 8.2. An illustration of the continuous uniform distribution with boundaries.

$$\mu = E[X] = \int_{-\infty}^{+\infty} xf(x)\,dx$$
$$= \int_a^b \frac{x}{b-a}\,dx = \frac{1}{b-a}\int_a^b x\,dx = \frac{1}{b-a} \times \frac{x^2}{2}\Big|_a^b = \frac{1}{b-a} \times \frac{b^2-a^2}{2}$$
$$= \frac{(b+a)(b-a)}{2(b-a)} = \frac{b+a}{2}$$

The variance of the $U(a,b)$ distribution is given by

$$\sigma^2 = \text{Var}(X) = E[(X-\mu)^2] = E[X^2] - \mu^2$$

In this case,

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f(x)dx = \int_a^b x^2 \frac{1}{b-a}dx = \frac{1}{b-a}\int_a^b x^2 dx = \frac{1}{b-a} \times \frac{x^3}{3}\Big|_a^b = \frac{b^3-a^3}{3(b-a)}$$
$$= \frac{(b-a)(b^2+ba+a^2)}{3(b-a)} = \frac{b^2+ba+a^2}{3}$$

Therefore,

$$\sigma^2 = \text{Var}(X) = E[X^2] - \mu^2 = \frac{b^2+ba+a^2}{3} - \left(\frac{b+a}{2}\right)^2$$
$$= \frac{4b^2+4ba+4a^2-3b^2-6ba-3a^2}{12} = \frac{b^2-2ba+a^2}{12} = \frac{(b-a)^2}{12}$$

The standard deviation, σ, is thus $\sigma = \frac{b-a}{\sqrt{12}}$

Example 8.1. Let us say the waiting time at a bus stop is uniformly distributed between 1 and 12 minutes. This means that the probability of waiting for any given amount of time between 1 and 12 minutes is the same. For instance, the probability

of waiting for exactly 5 minutes is the same as the probability of waiting for exactly 10 minutes. Similarly, the probability of waiting for less than 5 minutes is the same as the probability of waiting for more than 5 minutes. Showcase this using a few examples of your own.

Solution

To calculate the probability of waiting for a certain amount of time, we can use the formula for the continuous uniform distribution. For example, the probability of waiting for less than 5 minutes can be calculated as follows:

$$P(1 < X < 5) = \int_1^5 \frac{1}{b-a}\,dx = \frac{1}{b-a}x\Big|_1^5 = \frac{5-1}{12-1} = \frac{4}{11} \approx 0.36$$

This means that there is a 36% chance of waiting for less than 5 minutes at the bus stop. Similarly, the probability of waiting for more than 10 minutes can be calculated as follows:

$$P(10 < X < 12) = \int_{10}^{12} \frac{1}{b-a}\,dx = \frac{1}{b-a}x\Big|_{10}^{12} = \frac{12-10}{12-1} = \frac{2}{11} \approx 0.18$$

This means that there is an 18% chance of waiting for more than 10 minutes at the bus stop.

Do it in **R**:

```
# Define the range for the uniform distribution
a <- 1  # Minimum time
b <- 12 # Maximum time

# Probability Density Function (PDF)
pdf <- 1 / (b - a)  # For a uniform distribution, PDF is constant
pdf

## [1] 0.09090909

# Example 1: Probability of waiting exactly 5 minutes
# In a continuous distribution, this is always 0
prob_exactly_5 <- 0
prob_exactly_5

## [1] 0

# Example 2: Probability of waiting less than 5 minutes
prob_less_than_5 <- punif(5, min = a, max = b)
prob_less_than_5

## [1] 0.3636364

# Example 3: Probability of waiting more than 5 minutes
prob_more_than_5 <- 1 - prob_less_than_5
prob_more_than_5

## [1] 0.6363636
```

```r
# Example 4: Probability of waiting between 3 and 8 minutes
prob_between_3_and_8 <- punif(8, min = a, max = b) - punif(3, min = a, max
= b)
prob_between_3_and_8

## [1] 0.4545455
```

8.4 Exponential Distribution

The exponential distribution is a probability distribution that describes the time between independent and identically distributed events occurring at a constant rate. It is often used to model the time until the first occurrence of an event in a Poisson process. The probability density function (p.d.f) of the exponential distribution is given by:

$$f(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

where λ is non-negative and is the rate of the Poisson process. The cumulative distribution function of the exponential distribution is given by:

$$F(t) = \begin{cases} 1 - e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

The mean of the exponential distribution is given by:

$$\begin{aligned} E[X] &= \int_{-\infty}^{+\infty} x f(x) dx \\ &= \int_{-\infty}^{+\infty} x\lambda e^{-\lambda t} dx = \lambda \left\{ -x \frac{e^{-\lambda t}}{\lambda} \Big|_0^{+\infty} + \int_0^{+\infty} \frac{e^{-\lambda x}}{\lambda} dx \right\} \\ &= \text{(Note: by integration by parts)} = \lambda \left\{ 0 + \frac{1}{\lambda} \int_0^{+\infty} e^{-\lambda x} dx \right\} = -\frac{e^{-\lambda x}}{\lambda} \Big|_0^{\infty} \\ &= \frac{1}{\lambda} \end{aligned}$$

i.e. the mean of the exponential distribution is $1/\lambda$, the reciprocal of the rate of the Poisson process. Thus, the average time between events in a Poisson process is $1/\lambda$.

The variance, σ^2, for the exponential distribution is given by:

$$\sigma^2 = \text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$$

Thus,

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f(x)dx$$

$$= \int_0^{+\infty} x^2 \lambda e^{-\lambda x} dx = \lambda \left\{ x^2 \frac{e^{-\lambda t}}{-\lambda}\Big|_0^{+\infty} + \int_0^{+\infty} 2x \frac{e^{-\lambda x}}{\lambda} dx \right\}$$

$$= \text{(Note: by integration by parts)} = \lambda \left\{ 0 + \frac{2}{\lambda} \int_0^{+\infty} x e^{-\lambda x} dx \right\}$$

$$= \frac{2}{\lambda} \int_0^{\infty} x \lambda e^{-\lambda x} dx = \frac{2}{\lambda} E[X] = \frac{2}{\lambda^2}$$

Therefore,

$$\sigma^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Thus, the standard deviation, σ, of the exponential distribution is $1/\lambda$.

8.5 Normal Distribution

Normal distribution is the most important continuous probability distribution which describes many phenomena that occur in nature, industry and research. It is also known as Gaussian distribution, in honor of Karl Gauss, a German mathematician. The normal distribution is defined by its mean (which is the peak of the distribution), and standard deviation (which determines the spread). Most of the data from a normal distribution can be expected to lie within 3 standard deviations of the mean.

$\mathbb{DF}$: A continuous random variable X having the bell-shaped distribution is called a normal random variable.

The equation for the probability distribution depends upon two parameters, the mean μ and the variance σ^2 and usually denoted by $N(x; \mu, \sigma^2)$.

$\mathbb{DF}$: The probability density function of the normal distribution is defined as

$$N(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \quad for \ -\infty < x < \infty$$

where $\pi = 3.14159 \ldots$, $e = 2.71828 \ldots$

The main properties of the normal distribution are

i. The mean, mean and mode occur at $x = \mu$
ii. The curve is symmetrical about a vertical axis through the mean μ.
iii. The total area under the curve is 1.

Do it in **R**:

```
#normal distribution function
normal<- function(x, mu, sigma){
  (1/(sigma*sqrt(2*pi)))*exp((-1/2)*((x - mu) / sigma)^2)

}
```

```
par(mar = c(6.2,8,2,1))
curve(normal(x, 8,1), from = 0, to = 30,ylab = "Density", col = "black",ce
x.lab=2, cex.axis =2, ylim=c(0,0.5), lwd=2)
curve(normal(x, 8,4),  ylab = "", add= TRUE, col = "red", lwd=2)
curve(normal(x, 15,4),  ylab = "", add= TRUE, col = "blue", lwd=2)

legend("topright", col=c("black", "red", "blue"), legend = c(expression(th
eta~" = 8   "~sigma~" = 1 "), expression(theta~" = 8   "~ sigma~" = 4 "),
expression(theta~" = 15 "~sigma~" = 4 ")), box.lty=0, cex=1.5, lty = 1, in
set=0.1, lwd=2)
```

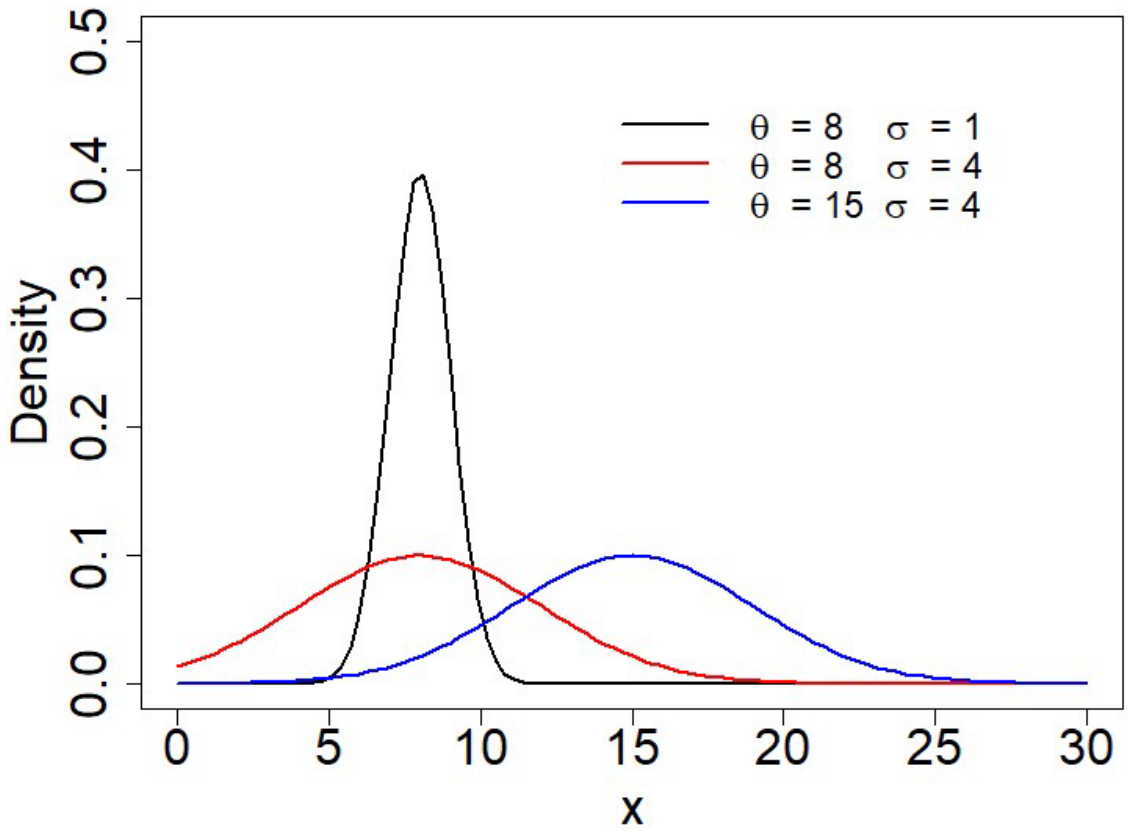


Figure 8.3. The black normal curve has the same mean as the red normal curve. Although the red curve has a bigger standard deviation. The red normal curve has a different mean from the blue normal curve but the same standard deviation. Black and blue normal curves have different means and standard deviations.

$\mathbb{DF}$: The distribution of a normal random variable with mean 0 and variance 1 is called a standard normal distribution.

We can transform any normal distribution to a standard one by means of transformation $Z = \frac{X-\mu}{\sigma}$. It follows that $X = Z\sigma + \mu$ and $dx = \sigma dz$

$$P(x_1 < X < x_2)$$
$$= \frac{1}{\sqrt{2\pi\sigma}} \int_{x_1}^{x_2} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{1}{2\sigma^2}(Z\sigma+\mu-\mu)^2} dx$$
$$= \frac{1}{\sqrt{2\pi} \times 1} \int_{z_1}^{z_2} e^{-\frac{1}{2\sigma^2}(Z-0)^2} dx = \int_{z_1}^{z_2} N(z; 0,1) dz = P(z_1 < Z < z_2)$$

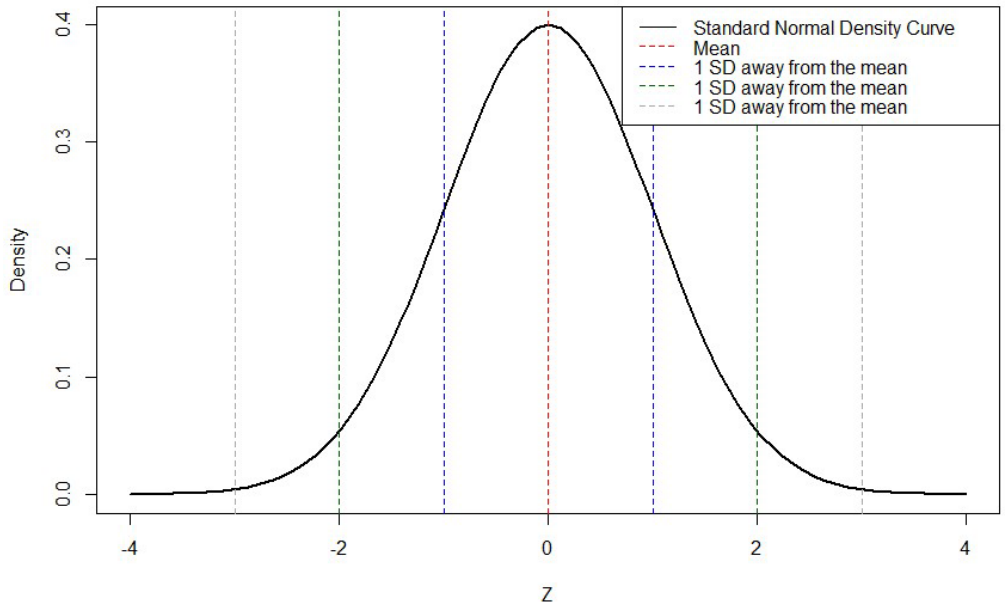


Figure 8.4. An illustration of the standard normal distribution. The vertical red dashed line represents the mean. The vertical blue dashed line represents 1 standard deviation from the mean, the vertical green dashed line represents 2 standard deviations from the mean and the vertical grey dashed line represents 3 standard deviations from the mean.

Example 8.2. The scores on a standardized test follow a normal distribution with a mean (μ) of 75 and a standard deviation (σ) of 10. A student named Alex scored 85 on the test. Calculate the z-score for Alex's score. Interpret the z-score in the context of the distribution.

Solution

To calculate the z-score, you can use the formula:

$$z = \frac{X - \mu}{\sigma}$$

where X is the individual score (85 for Alex), μ is the mean (75) and σ is the standard deviation (10). Now calculating the z-score for Alex:

$$z = \frac{85 - 75}{10} = 1$$

A z-score of 1 means that Alex's score is 1 standard deviation above the mean. In the context of the distribution, this indicates that Alex performed relatively well compared to the average student, as his score is above the mean.

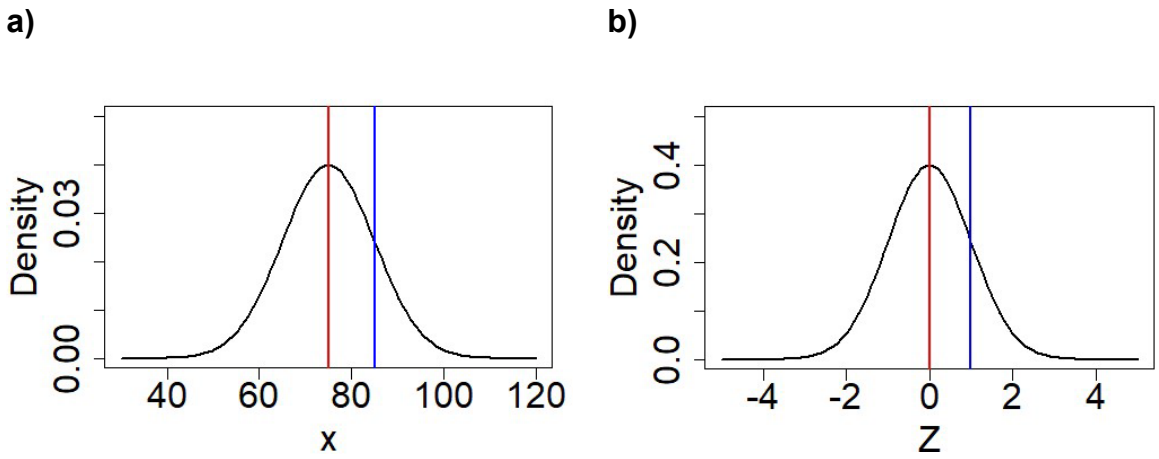


Figure 8.5. An illustration of a normal distribution (Panel a)) and corresponding transformation to a standard normal distribution (Panel b)).

9 Means and Variances of Linear Combination of Random Variables along with Covariance

9.1 Learning Outcomes

At the end of this section, you must be able to:

- Define random variables and linear combinations of random variables.
- Understand the concept of means and variances of random variables.
- Express the mean of a linear combination of random variables using the appropriate mathematical notation.
- Express the variance of a linear combination of random variables using the appropriate mathematical notation.
- Recognise the relationship between the covariance of random variables and the variance of linear combinations.

9.2 Introduction

When dealing with random variables, it's common to encounter linear combinations of these variables. A linear combination is a mathematical expression that involves the sum and/or scalar multiplication of random variables. Understanding the means and variances of linear combinations is crucial in various fields, including statistics and probability theory.

9.3 Expectation (mean) of linear combination of random variables

$\mathbb{TH}$: For any random variables X and Y

$$E[X+Y] = E[X] + E[Y]$$

It follows that if we have n random variables $X_1, X_2, \dots, X_n$ then:

$$E\left[\sum_{i=1}^{n} X_i\right] = E[X_1, X_2, \dots, X_n] = E[X_1] + E[X_2] + \dots + E[X_n] = \sum_{i=1}^{n} E[X_i]$$

Since we know that $E[aX] = aE[X]$ then $E[X-Y] = E[X] - E[Y]$

$\mathbb{TH}$: Linearity of expectation

For n random variables $X_1, X_2, \dots, X_n$ and constants $a_1, a_2, \dots, a_n$

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

The great thing about these properties is that no independence is required.

Example 9.1. What is the eexpected value of X for two fair dice?

The p.d.f for one die is:

X	1	2	3	4	5	6
$P(X=x)$	1/6	1/6	1/6	1/6	1/6	1/6

$$E[X] = \sum_{i=1}^{6} x_i f(x_i) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$$
$$= \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3\frac{1}{2}$$

The same is true for the second die. Hence $E[X+Y] = 3\frac{1}{2} + 3\frac{1}{2} = 7$

We did not have to assume that dice are independent. The expected sum is 7 even if they are glued together (providing that gluing does not make each individual die unfair).

9.4 Covariance

The extent to which two random variables vary together can be measured by there covariance. Let us consider two random variables, X and Y, having means μ_X and μ_Y respectively.

$\mathbb{DF}$: Covariance is defined as:

$$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Alternatively, this can be defined as:

$$\text{Cov}(X,Y) = E[XY] - E[X] \times E[Y]$$

Proof

$$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y]$$
$$= E[XY] - \mu_X E[Y] - \mu_Y E[X] + \mu_X \mu_Y = E[XY] - \mu_X \mu_Y$$

i.e. $\text{Cov}(X,Y) = E[XY] - E[X] \times E[Y]$

Properties of Covariance

i. $\text{Cov}(aX, bY) = ab\text{Cov}(X,Y)$

ii. $\text{Cov}(X+a, Y+b) = \text{Cov}(X,Y)$

iii. $\text{Cov}(X,X) = \text{Var}(X)$

iv. If X and Y are independent, then $\text{Cov}(X,Y) = 0$

This follows immediately from the fact shown earlier that if X and Y are independent then $E[XY] = E[X] \times E[Y]$. We should note however that the

converse is not necessarily true, i.e. if $E[XY] = E[X] \times E[Y]$ then X and Y are not necessarily independent.

v. $\text{Cov}(X+Y, U+V) = \text{Cov}(X,U) + \text{Cov}(X,V) + \text{Cov}(Y,U) + \text{Cov}(Y,V)$

Positive covariance indicates that higher than average values of one variable tend to be paired with higher than average values of the other variable.

Negative covariance indicates that higher than average values of one variable tend to be paired with lower than average values of the other variable.

9.5 Variance of Linear Combination of Random Variables

The variance of linear combination of random variables are given by:

i. $\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2)$

ii. $\text{Var}(X_1 - X_2) = \text{Var}(X_1) + \text{Var}(X_2) - 2\text{Cov}(X_1, X_2)$

Proof

$$\begin{aligned}\text{Var}(X_1 - X_2) &= E[\{(X_1 - \mu_1) - (X_2 - \mu_2)\}^2] \\ &= E[(X_1 - \mu_1) - 2(X_1 - \mu_1)(X_2 - \mu_2) + (X_2 - \mu_2)^2] \\ &= \text{Var}(X_1) - 2\text{Cov}(X_1, X_2) + \text{Var}(X_2)\end{aligned}$$

It follows that if X_1 and X_2 are independent then:

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2)$$

and

$$\text{Var}(X_1 - X_2) = \text{Var}(X_1) + \text{Var}(X_2)$$

We can state more generally that:

$$\text{Var}(a_1X_1 + a_2X_2) = a_1^2\text{Var}(X_1) + a_2^2\text{Var}(X_2) + 2a_1a_2\text{Cov}(X_1, X_2)$$

and even more generally as follows:

$$\text{Var}\left(\sum_i a_iX_i\right) = \sum_i a_i^2\text{Var}(X_i) + 2\underbrace{\sum\sum a_ia_j\text{Cov}(X_i, X_j)}_{i<j}$$

This last expression can also be written as:

$$\text{Var}\left(\sum_i a_iX_i\right) = \sum_i\sum_j a_ia_j\text{Cov}(X_i, X_j)$$

or even

$$\text{Var}\left(\sum_i a_iX_i\right) = \sum_i a_i^2\text{Var}(X_i) + 2\underbrace{\sum\sum a_ia_j\text{Cov}(X_i, X_j)}_{i\neq j}$$

More specifically if X_i's are jointly independent of each other then:

$$\text{Var}\left(\sum_i a_i X_i\right) = \sum_i a_i^2 \text{Var}(X_i)$$

If the a_i's are all equal to the value, $1/n$ then we have

$$Var(\bar{X}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \text{Var}\left(\sum_{i=1}^{n}\frac{1}{n}X_i\right) = \frac{1}{n^2}\text{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\left(\sum_{i=1}^{n}\text{Var}(X_i)\right)$$
$$= \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}$$

9.6 Sums of independent random variables from known distributions

Sums of independent random variables from a common family of distributions are often also from the same family. Using the results for the expectations and the variance of sums of random variables the following examples are given for the Binomial (b), Poisson (p), Normal (N), Geometric (g) and Negative Binomial (NB) distributions:

$$b(n_1, p) + b(n_2, p) = b(n_1 + n_2, p)$$

$$p(\mu_1) + p(\mu_2) = p(\mu_1 + \mu_2)$$

$$N(\mu_1, \sigma_1^2) + N(\mu_2, \sigma_2^2) = N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$g(p) + g(p) = NB(2, p)$$

These formulae obviously extend to sums of more than two random variables.

9.7 Distributions of the sums of certain random variables

i. If $X_1, X_2, \ldots, X_n$ are iid (independently and identically distributed, i.e. each has the same probability distribution as the others and all are mutually independent) Bernoulli random variables with probability of success p and probability of failure $(1 - p)$ then

$$\sum_{i=1}^{n} X_i \sim b(n, p)$$

ii. If $X_1, X_2, \ldots, X_n$ are independent Poisson random variables with X_i having mean μ_i then:

$$\sum_{i=1}^{n} X_i \sim \text{Poisson distribution with mean } \sum_{i=1}^{n} \mu_i$$

iii. If $X_1, X_2, \ldots, X_n$ are independent normal random variables with probability distributions, $X_i \sim N(\mu_i, \sigma_i^2)$ then:

$$a_i X_i \sim N(a_i \mu_i, a_i^2 \sigma_i^2)$$

or more generally,

$$\sum_{i=1}^{n} a_i X_i \sim N\left(\sum_{i=1}^{n} a_i \mu_i\,, \sum_{i=1}^{n} a_i^2 \sigma_i^2\right)$$

In particular, if $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ and X and Y are independent then:

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

$$X - Y \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

If $X_1, X_2, \ldots, X_n$ are iid random variables with probability distribution, $X_i \sim N(\mu, \sigma^2)$ then:

$$X_1, X_2, \ldots, X_n \sim N(n\mu, n\sigma^2)$$

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Example 9.2. The random variables X_1, X_2 and X_3 have the following means and variances

	X_1	X_2	X_3
Mean	2	-1	4
Variance	4	6	8

and covariances

$\text{Cov}(X_1, X_2) = 1$ $\text{Cov}(X_1, X_3) = -1$ $\text{Cov}(X_2, X_3) = 0$

What is the mean and the variance of $Y = 3X_1 + 4X_2 - 6X_3$?

Solution

$$E[Y] = E[3X_1 + 4X_2 - 6X_3] = 3 \times 2 + 4 \times (-1) - 6 \times 4 = -22$$

$$\begin{aligned} Var(Y) &= Var(3X_1 + 4X_2 - 6X_3) \\ &= 3^2 \times 4 + 4^2 \times 6 + (-6)^2 \times 8 \\ &\quad + 2 \times [3 \times 4 \times 1 + 3 \times (-6) \times (-1) + 4 \times (-6) \times (-1) + 4 \times (-6) \times 0] \\ &= 480 \end{aligned}$$

10 Moment generating function (MGF)

10.1 Learning Outcomes

At the end of this section, you must be able to:

- Understand the definition of the moment generating function as a mathematical function associated with a random variable.
- Recognise the MGF as a tool for characterising the distribution of a random variable.
- Apply MGFs to solve problems in probability and statistics.

10.2 Introduction

The purpose of a moment generating function is to determine moments of random variables. However, the most important contribution of the moment generating functions is to establish distribution of functions of random variables.

If $u(X) = X^r$ for $r = 0,1,2, \ldots$ then the definition of the rth moment about the origin of the random variable X, which we introduced earlier and denoted by μ_r', can be written as follows:

$$\mu_r' = E[X^r] = \begin{cases} \sum_{\text{all } x} x^r f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} x^r f(x)\, dx & \text{if } X \text{ is continuous} \end{cases} \qquad (*)$$

Since the first and the second moments about the origin are given by $\mu_1' = E[X]$ and $\mu_2' = E[X^2]$, we can define the mean and the variance as $\mu = \mu_1'$ and $\sigma^2 = \mu_2' - \mu^2 = \mu_2' - (\mu_1')^2$. Although the higher moments of a random variable can be determined directly from

the definition (*), an alternative procedure exists. This procedure requires a definition

of moment generating function (MGF).

$\mathbb{DF}$: The moment generating function of the random variable X with p.d.f $f(x)$ (if it exists) is given by and is denoted by $E[e^{tX}]$ and is denoted by $M_X(t)$. Hence,

$$M_X(t) = \begin{cases} \sum_{all\ x} e^{tx} f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} e^{tx} f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Note that this definition bears a striking similarity to the definition of a Laplace transform i.e. $\int_0^{\infty} e^{-tx} \times f(x)dx$

If a moment generating function of a random variable does exist, it can be used to generate all the moments of that variable. The method is described as follows.

$\mathbb{TH}$: Let X be a random variable with moment generating function $M_X(t)$. Then,

$$\mu_r' = \left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0}$$

Proof (discrete case)

If $M_X(t) = \sum_{\text{all } x} e^{tx} f(x)$ then

(1) $M_X(0) = \sum_{\text{all } x} e^0 f(x) = \sum_{\text{all } x} f(x) = 1$

(2) $M'_X(t) = \frac{dM_X(t)}{dt} = \sum_{\text{all } x} xe^{tx} \times f(x)$

$\therefore M'_X(0) = \left.\frac{dM_X(t)}{dt}\right|_{t=0} = \sum_{\text{all } x} xe^0 \times f(x) = \sum_{\text{all } x} xf(x) = \mu_1' = E[X]$

(3) $M_X^r(t) = \left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0} = \sum_{\text{all } x} x^r e^{tx} \times f(x)$

$\therefore M_X^r(0) = \left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0} = \sum_{\text{all } x} x^r e^0 \times f(x) = \sum_{\text{all } x} x^r f(x) = \mu_r' = E[X^r]$

Thus, to find the rth moment about the origin we simply find the rth derivative of the MGF and then equate t to 0. This property of the moment generating function is true for all random variables, both discrete and continuous, for which the MGF exists.

Example 10.1. Find the moment generating function of the Binomial random variable X

and then use it to verify that $\mu = np$ and $\sigma^2 = npq$.

Solution

The p.d.f for the Binomial distribution is given by:

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} = \binom{n}{x} p^x q^{n-x} \quad for\ x = 0,1,2,\dots,n$$

$$M_X(t) = \sum_{all\ x} e^{tx} f(x) = \sum_{all\ x} e^{tx} \binom{n}{x} p^x q^{n-x} = \sum_{all\ x} \binom{n}{x} (pe^t)^x q^{n-x}$$

The sum is a binomial expansion of $(pe^t + q)^n$ and therefore $M_X(t) = (pe^t + q)^n$

$$\frac{dM_X(t)}{dt} = n(pe^t + q)^{n-1} pe^t$$

Setting $t = 0$

$$\mu_1' = \mu = n(pe^0 + q)^{n-1} pe^0 = n(p+q)^{n-1} p = np \text{ (since } p + q = 1)$$

$$\frac{d^2 M_X(t)}{dt^2} = npe^t (n-1)(pe^t + q)^{n-2} pe^t + n(pe^t + q)pe^t$$
$$= np[e^t (n-1)(pe^t + q)^{n-2} pe^t + (pe^t + q)e^t]$$

Setting $t = 0$

$$\mu_2' = np[(n-1)p+1]$$

$$\sigma^2 = \mu_2' - (\mu_1')^2 = np[(n-1)p+1] - n^2p^2 = n^2p^2 - np^2 + np - n^2p^2 = pn(1-p) = npq$$

10.3 The uniqueness property

$\mathbb{TH}$: Let X and Y be two random variables with MGF $M_X(t)$ and $M_Y(t)$ respectively. If $M_X(t) = M_Y(t)$ for all values of t then X and Y have the same probability distribution.

10.4 Other properties

Looking ahead, we can show that the MGF of the sum of two independent random variables is the product of the two individual MGFs. The joint p.d.f of two independent random variables follows immediately from the multiplication rule for independent events that

$$P[(X = x) \cap (Y = y)] = P(X = x) \times P(Y = y)$$

Therefore,

$$M_{X+Y}(t) = E[e^{tX+tY}] = E[e^{tX} \times e^{tY}] = E[e^{tX}] \times E[e^{tY}]$$

i.e. $M_{X+Y}(t) = M_X(t)M_Y(t)$ if X and Y are independent. This may be used to show, amongst other things, that:

i. The sum of Poisson random variables is Poisson
ii. The sum of Normal random variables is Normal
iii. The sum of exponential random variables is gamma
iv. The sum of gamma random variables is gamma
v. The sum of χ^2 random variables is χ^2

Example 10.2. If we consider the following p.d.f

x	0	1	3	5
$f(x)$	0.3	0.4	0.15	0.15

then the MGF is simply

$$M_X(t) = \sum_{\text{all } x} e^{tx} f(x) = e^{0t} \times 0.3 + e^{1t} \times 0.4 + e^{3t} \times 0.15 + e^{5t} \times 0.15$$

Example 10.3. Find the MGF of the following distribution and use it to determine the mean and variance

$$f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Solution

The MGF is

$$M_X(t) = \int_0^\infty e^{tx}e^{-x}dx = \int_0^\infty e^{-x(1-t)}dx = \frac{e^{-x(1-t)}}{-(1-t)}\Big|_0^\infty = \frac{1}{1-t} \qquad t < 1$$

We can show that $M_X^1(t) = (1-t)^{-2}$ and $M_X^2(t) = 2(1-t)^{-3}$. Thus,

$$E[X] = M_X^1(t)|_{t=0} = 1$$

and

$$E[X^2] = M_X^2(t)|_{t=0} = 2.$$

Also,

$$\mathrm{Var(X)} = E[X^2] - \{E[X]\}^2 = 2 - 1 = 1$$

Example 10.4. A random variable X has a discrete uniform distribution given by

$$f(x,n) = \begin{cases} \frac{1}{n} & x = 1,2,\dots,n \\ 0 & \text{elsewhere} \end{cases}$$

Show that the MGF of X is

$$M_X(t) = \frac{e^t(1-e^{nt})}{n(1-e^t)}$$

Solution

$$M_X(t) = \sum_{x=1}^{n} e^{tx}\frac{1}{n} = \frac{1}{n}\sum_{x=1}^{n} e^{tx} = \frac{e^t}{n}\left(1 + e^t + e^{2t} + \cdots + e^{(n-1)t}\right)$$

The sum in the parenthesis is the sum of elements of geometric progression sum $= \frac{(1-r^n)}{(1-r)}$ where $r = e^t$ so

$$M_X(t) = \frac{e^t}{n} \times \frac{(1-e^{nt})}{(1-e^t)} = \frac{e^t(1-e^{nt})}{n(1-e^t)}$$

11 Parameter estimation

11.1 Learning Outcomes

At the end of this section, you must be able to:

- Understanding the concept of statistical inference
- Understanding the properties of a good estimator

11.2 Introduction

In the lectures which follow we will consider problems of Statistical Inference. Statistical inference is based upon the logical process of inductive inference. Progress in further expanding our knowledge often rises out of observing some situation in detail. We observe a given situation and collect data which describes what we observe. On the basis of the data itself we draw certain conclusions. However, the conclusions themselves usually go further than the actual observed situations, i.e. we try to generalise from the particular situation we observed to all situations which are similar to the one we observed. For example, we went to a particular T-junction and waited to see how many cars out of a total of 200 turn left. Suppose we saw 68 cars out of 200 turn left. Therefore the observed proportion of cars that turned left was 68/200=0.34. What we would now like to say is that the proportion of all cars approaching the T-junction that turn left is 'close to' 0.34. This sort of extension from the particular to the general is described formally as inductive inference. Clearly inductive inference can be a hazardous process. Is our estimate of the proportion of all cars that turn left at the T-junction a good one? How 'close to' the true value is it? Obviously inductive inference carries with it some degree of uncertainty – we simply cannot make hard and fast generalization on the basis of the limited information we have collected. However, inferences can be made with the aid of statistical theory that involves a measure of uncertainty in their description. For a formal description of uncertainty we use Probability Theory and so that was why we spent the time we did on consideration of the Probability Theory.

11.3 The target population, a sample and simple random sampling

Before we proceed to the problem of inference proper, we need to frequent ourselves with some terminology.

Suppose we wanted to know what the average weight of staff in the University was. Then the target population in this case consists of all staff in the University. Clearly it would be very time consuming (and perhaps even impossible) to measure every staff member's weight and so we are more or less forced into taking a sample from the target population. We choose a group of staff within the University and measure their weights. On the basis of this information we try to infer something about the average weight of all staff at the University.

But how should the sample be chosen? Clearly we don't want to choose staff who are all incredibly heavy or incredibly light. One of the fundamental assumptions of most statistical analyses is that data is representative of the population as a whole – that a sample is a random sample. In order to obtain a simple random sample

a) Each member of the population is identified. For example, the staff in the University is identified via the records kept by the Human Resources Department.
b) Each member of the population is given an equal chance of being chosen as a member of the sample and members chosen independently of one other. Continuing the above example, suppose there are 940 members of staff in the University and we wanted a sample of 10 staff. Each staff member could be assigned a number between 1 and 940 and then a table of random numbers used to obtain the numbers corresponding to the 10 staff members to be included in the sample.

11.4 Parameter estimation

One of the primary roles of Statistical Inference is to provide estimates of population parameters on the basis of information supplied by a sample. For example, we might have obtained the following weights (in kgs) for the 10 people in our simple random sample:

51.3 76.4 57.3 89.1 52.1 70.3 62.4 63.0 95.1 86.1

What then would be our estimate of the population mean, μ, and the standard deviation, σ, on the basis of this information?

One obvious estimator for the population mean, μ, is the sample mean, $\bar{X}$, where

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Using this estimator, $\mu = \bar{X}$, our estimate of μ would be

$$\frac{51.3 + 76.4 + 57.3 + 89.1 + 52.1 + 70.3 + 62.4 + 53.0 + 95.1 + 86.1}{10} = \frac{703.1}{10} = 70.31$$

Also, we could use the sample standard deviation S as our estimator for the population standard deviation σ. Using this estimator, $\hat{\sigma} = S$, our estimate of σ would be 15.744. However, there are always other estimators we could use. For example, in order to estimate μ we could use

a) the sample median,

b) the average of the 2nd and 8th ordered values,

c) the average of the minimum and maximum values. etc.

Clearly, there are indefinitely many estimators that we could use and this is true for any parameter we care to choose!

11.5 Properties of good estimators

How then should we decide on our choice of estimator? There are several properties that we should like our choice of estimator to exhibit. Letting θ denote the parameter of interest and $\hat{\theta}$ our estimator of it, these are:

11.5.1 Unbiasedness

$\mathbb{DF}$: A statistic $\hat{\theta}$ is an unbiased estimator of the parameter θ iff $E[\hat{\theta}] = \theta$

The quantity $E[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

11.5.2 Consistency

The property means that if the whole population were eventually to be observed then the estimator should give the correct answer.

$\mathbb{DF}$: A statistic $\hat{\theta}$ is a (weakly) consistent estimator of θ if for each positive constant c, $\lim_{n\to\infty} P(|\hat{\theta} - \theta| \geq c) = 0$

$\mathbb{TH}$: (Chebyshev's Theorem)

If the random variable X has mean μ and variance σ^2 then for any constant $k > 0$

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

or

$$P(|X - E[X]| \geq k\sigma) \leq \frac{\text{Var}(X)}{c^2}$$

The following Theorem is a consequence of Chebyshev's Theorem and can help us to

determine whether the estimator is consistent.

$\mathbb{TH}$: If $\hat{\theta}$ is unbiased and $\lim_{n\to\infty} \text{Var}(\hat{\theta}) = 0$ then the statistic $\hat{\theta}$ is consistent estimator of θ.

11.5.3 Sufficiency

An estimator $\hat{\theta}$ is said to be sufficient for the parameter θ if it is summarises all the

information contained in a random sample relevant to the estimation of the population

parameter θ

$\mathbb{DF}$: (the factorization criterion)

The statistic $\hat{\theta}$ is sufficient estimator for θ iff, for each value of $\hat{\theta}$, the conditional distribution of the random sample $x_1, x_2, \ldots, x_n$ given $\hat{\theta}$, is independent of θ. The factorization criterion is:

$\hat{\theta}$ is a sufficient statistic for θ iff $f(x|\theta) = u(x)v(\hat{\theta}, \theta)$ where the functions u and v are both non-negative.

11.5.4 Relative efficiency

$\mathbb{DF}$: If $\hat{\theta}_1$ and $\hat{\theta}_2$ are two unbiased estiamtors of a parameter θ and $Var(\hat{\theta}_1) \leq Var(\hat{\theta}_2)$ then $\hat{\theta}_1$ is relatively more efficient then $\hat{\theta}_2$.

If we consider all possible unbiased estimators of some parameter θ, the one with the smallest variance is called the most efficient estimator of θ.

Example 11.1

X_1 and X_2 are independent random variables drawn from a population with mean μ and variance σ^2, i.e. $E[X_1] = E[X_2] = \mu$ and $Var(X_1) = Var(X_2) = \sigma^2$. Suppose we want to estimate μ on the basis of the sample of these two random variables. Let's consider the following three estimators:

1) $T_1 = \frac{1}{2}X_1 + \frac{1}{2}X_2$

2) $T_2 = \frac{1}{3}X_1 + \frac{2}{3}X_2$

3) $T_3 = \frac{1}{5}X_1 + \frac{2}{5}X_2$

Solution

For T_1 we have

$$E[T_1] = E\left[\frac{1}{2}X_1 + \frac{1}{2}X_2\right] = E\left[\frac{1}{2}X_1\right] + E\left[\frac{1}{2}X_2\right] = \frac{1}{2}E[X_1] + \frac{1}{2}E[X_2]$$

(Since $E[aX] = aE[X]$)

$$= \frac{1}{2}\mu + \frac{1}{2}\mu = \mu$$

i.e. T_1 is unbiased. Also,

$$\text{Var}[T_1] = \text{Var}\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \text{Var}\left(\frac{1}{2}X_1\right) + \text{Var}\left(\frac{1}{2}X_2\right)$$

(as X_1 and X_2 are independent)

$$= \left(\frac{1}{2}\right)^2 \text{Var}(X_1) + \left(\frac{1}{2}\right)^2 \text{Var}(X_2)$$

(since $\text{Var}(aX) = a^2\text{Var}(X)$)

$$= \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 = 0.5\sigma^2$$

For T_2 we have

$$E[T_2] = E\left[\frac{1}{3}X_1 + \frac{2}{3}X_2\right] = E\left[\frac{1}{3}X_1\right] + E\left[\frac{2}{3}X_2\right] = \frac{1}{3}E[X_1] + \frac{2}{3}E[X_2]$$

(Since $E[aX] = aE[X]$)

$$= \frac{1}{3}\mu + \frac{2}{3}\mu = \mu$$

i.e. T_2 is unbiased. Also,

$$\text{Var}[T_2] = \text{Var}\left(\frac{1}{3}X_1 + \frac{2}{3}X_2\right) = \text{Var}\left(\frac{1}{3}X_1\right) + \text{Var}\left(\frac{2}{3}X_2\right)$$

(as X_1 and X_2 are independent)

$$= \left(\frac{1}{3}\right)^2 \text{Var}(X_1) + \left(\frac{2}{3}\right)^2 \text{Var}(X_2)$$

(since $\text{Var}(aX) = a^2\text{Var}(X)$)

$$= \frac{1}{9}\sigma^2 + \frac{4}{9}\sigma^2 = \frac{5}{9}\sigma^2 = 0.556\sigma^2$$

For T_3 we have

$$E[T_3] = E\left[\frac{1}{5}X_1 + \frac{2}{5}X_2\right] = E\left[\frac{1}{2}X_1\right] + E\left[\frac{2}{5}X_2\right] = \frac{1}{5}E[X_1] + \frac{2}{5}E[X_2]$$

(Since $E[aX] = aE[X]$)

$$= \frac{1}{5}\mu + \frac{2}{5}\mu = \frac{3}{5}\mu$$

i.e. T_3 is not unbiased. Also,

$$\text{Var}[T_3] = \text{Var}\left(\frac{1}{5}X_1 + \frac{5}{5}X_2\right) = \text{Var}\left(\frac{1}{5}X_1\right) + \text{Var}\left(\frac{2}{5}X_2\right)$$

(as X_1 and X_2 are independent)

$$= \left(\frac{1}{5}\right)^2 \text{Var}(X_1) + \left(\frac{2}{5}\right)^2 \text{Var}(X_2)$$

(since $\text{Var}(aX) = a^2\text{Var}(X)$)

$$= \frac{1}{25}\sigma^2 + \frac{4}{25}\sigma^2 = \frac{5}{25}\sigma^2 = \frac{1}{5}\sigma^2 = 0.2\sigma^2$$

Therefore, of the three estimators considered here, only T_1 and T_2 are unbiased. In addition, T_1 is preferable to T_2 since $\text{Var}(T_1) < \text{Var}(T_2)$.

11.6 Estimating the population mean

The most commonly used estimator of a population mean, μ , is the sample mean

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

where $[X_1, X_2, \dots, X_n]$ is a sample of n independent values from a population with mean μ and variance σ^2.

Now,

$$E[\bar{X}] = E\left[\frac{\sum_{i=1}^{n} X_i}{n}\right] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n} n\mu = \mu$$

i.e. $\bar{X}$ is an unbiased estimator of μ. Also,

$$\text{Var}[\bar{X}] = \text{Var}\left[\frac{\sum_{i=1}^{n} X_i}{n}\right] = \frac{1}{n^2}\sum_{i=1}^{n} \text{Var}[X_i] = \frac{\sigma^2}{n}$$

Thus, $\bar{X}$ is consistent as $\lim_{n\to\infty}[\text{Var}(\bar{X})] = 0$

11.7 Estimation of a population proportion *p*

Suppose we wish to estimate a population proportion, p. For example, we might want to know what proportion of people in Nottingham are under 172 cm in height. Suppose we were to take a sample of n people living in Nottingham and found that X of them were under 172 cm in height. Thus, X represents the number of observed "successes", a "success" being in this case a person having a height less than 172 cm. Clearly X is a random variable, because if we took another sample of size n we would expect the value of X to vary between the two samples. The probability of any one chosen person being under 172 cm is given by p, i.e. the probability of a "success" is p. Because we are taking a simple random sample the peoples' heights are independent, i.e. trials are independent. By definition, we have n trials. It therefore follows that X is a Binomial random variable with mean np and variance npq.

The most commonly used estimator of a population proportion p is $\hat{p} = \frac{X}{n}$, the sample proportion, where X is the number of "successes" observed and n is the sample size, or number of trials. For this estimator we have

$$E\left[\hat{P}\right] = E\left[\frac{X}{n}\right] = \frac{1}{n}E[X] = \frac{1}{n}np = p$$

i.e. $\hat{P}$ is unbiased. Also,

$$\text{Var}\left(\hat{P}\right) = \text{Var}\left(\frac{X}{n}\right) = \frac{1}{n^2}\text{Var}(X) = \frac{1}{n^2}npq = \frac{pq}{n}$$

Thus, $\hat{P}$ is consistent as $Var\left(\hat{P}\right) \to 0$ as $n \to \infty$.

12 Method of moments (MOM)

12.1 Learning Outcomes

At the end of this section, you must be able to:

- Define the method of moments and its fundamental principles.
- Explain the concept of moments and their role in statistical estimation.

12.2 Introduction

The method of moments is a powerful statistical technique used for estimating parameters in probability distributions. Rooted in the concept of moments, this method provides a systematic approach to derive estimators for population parameters based on sample moments. Moments, in this context, capture key features of a distribution, such as mean, variance, skewness, and kurtosis.

12.3 Moments of a distribution and samples moments

We recall that the rth moment of a distribution is given by

$$\mu_r' = E[X^r] = \begin{cases} \sum_R x^r f(x) & \text{if } X \text{ is discrete} \\ \int_R x^r f(x)dx & \text{if } X \text{ is continuous} \end{cases}$$

Note, that we can similarly define central moments as $\mu_r = E[(X - \mu')^r]$. Typically, a moment, be it a raw or a central moment will be a function of the unknown parameters of the distribution and so for the rth moment of a distribution we can write $\mu_r' = g(\theta_1, \dots, \theta_k)$.

$\mathbb{DF}$: For a sample of observations $X_1, X_2, \dots, X_n$ taking values $x_1, x_2, \dots, x_n$ the rth raw moment is:

$$M_r' = \frac{\sum X^r}{n}$$

12.4 Method of Moments Estimation

This method simply states that to estimate k parameters of a distribution we equate the first k sample moments to their theoretical moments and solve these k simultaneous equations for the parameter estimates.

Let a random variable X has a p.d.f that are of known functional form but depends upon unknown parameter θ which may take any value from the set Θ. The set Θ is called the parameter space. Thus, we are dealing not with one but with the family of distributions. For each value of $\theta \in \Theta$ there is one member of the family denoted $\{f(x;\theta);\ \theta \in \Theta\}$. The family can be defined by several parameters $\theta_1, \dots, \theta_k$. In this case we are looking for a member of the family $f(x; \theta_1, \dots, \theta_k)$.

Example 12.1. We wish to estimate the parameter k in the distribution $f(x;k) = (k+1)x^k$ for $0 < x < 1$ and $k > -1$. We also have a sample of 8 values 0.5, 0.2, 0.9, 0.8, 0.8, 0.4, 0.7, 0.9. The first moment is

$$\mu_1' = \int_0^1 x(k+1)x^k dx = \int_0^1 (k+1)x^{k+1} dx = (k+1)\frac{x^{k+2}}{k+2}\Big|_0^1 = \frac{k+1}{k+2}$$

The first sample moment is:

$$M_1' = \bar{X} = \frac{0.5 + 0.2 + 0.9 + 0.8 + 0.8 + 0.4 + 0.7 + 0.9}{8} = 0.65$$

and if we equate the theoretical and sample moment, we get $k = \hat{k}$ where $\frac{\hat{k}+1}{\hat{k}+2} = \bar{X}$ i.e. $\hat{k} = \frac{2\bar{X}-1}{1-\bar{X}}$.

This is our Method of Moments estimator of k. The Method of Moments estimate is obtained by replacing $\bar{X}$ by its value $\bar{x} = 0.65$, giving the estimate for k such that $\hat{k} = 0.857$.

Example 12.2. (The Gamma Distribution) Given a random sample $X_1, X_2, \ldots, X_n$ from the gamma distribution:

$$f(x;\theta,k) = \frac{1}{\theta^k \Gamma(k)} x^{k-1} e^{-\frac{x}{\theta}}$$

derive the Method of Moments estimators for both θ and k.

Solution

We can show that the first two moments are:

$$\mu_1' = k\theta$$

$$\mu_2' = k(k+1)\theta^2 \quad (\mu_2 = k\theta^2)$$

so, we put:

$$\hat{k}\hat{\theta} = \bar{X}$$

$$\hat{k}\hat{\theta}^2 = \text{Var}(X)$$

These are solved to give:

$$\hat{\theta} = \frac{\text{Var}(X)}{\bar{X}}$$

$$\hat{k} = \frac{\bar{X}^2}{\text{Var}(X)}$$

12.5 Comments on the Method of Moments estimators

It is unfortunate that such a simple concept as Method of Moments Estimation does not lead to estimators with particularly favourable properties. All that can be said for

them is that they are consistent estimators. In general, they are not unbiased nor are they efficient (not even asymptotically so).

However, they are a useful illustration of a method of estimation, and on occasion provide more easily evaluated estimators than, say, Maximum Likelihood Estimators. The Gamma distribution example above is one such case. The evaluation of Method of Moments Estimators is not beset with problems when the range depends on the parameter as in the Uniform distribution nor when the distribution is not continuous. It will even work when the distribution is a mixture of continuous and discrete. An example of this latter case is the time one must wait at traffic lights, when there is a discrete probability that the time will be zero, and a truncated exponentially distributed time of wait if it is not zero.

Additionally, the following points may be made:

i. If we use central rather than raw moments, the estimators are unaltered. That is to say that the values of the estimators would be exactly the same whether raw or central moments or a combination.
ii. In the setting out of the method we stated that we equate the first k sample and theoretical moments. If one of the moments is independent of the parameters, then we will have to use a different set of k moments. For example, a symmetrical distribution centred on zero will have a mean μ_1' of zero.
iii. Similarly, if one of the theoretical moments is a direct function of another then the first k moments will not be independent. We can easily modify the definition to refer to the first suitable k moments to deal with the problems set up in this and the previous note.
iv. Taking the argument further there is no theoretical reason against using any k independent moments. It may be that the set of other than the first independent k moments will yield equations that are more tractable to manipulation and solution. However, this raises the possibility (certainty) of different Method of Moments Estimators of the same parameters even when using the same data. As a simple example we may wish to estimate the parameter μ in a Poisson distribution $p(\mu)$. If our data have a mean of 1.2 and a variance of 1.5 then we could use either of these values as the estimate of the parameter μ.

13 Maximum Likelihood Estimates

13.1 Learning Outcomes

At the end of this section, you must be able to:

- Understand the principle of Maximum Likelihood Estimation and how the likelihood function is used to estimate model parameters based on observed data.
- Be able to derive the Maximum Likelihood Estimates by formulating the likelihood function and solving for the parameter values that maximize it.
- Apply Maximum Likelihood Estimation to estimate parameters for various statistical models and use software tools to compute and interpret these estimates.

13.2 Introduction

The Method of Maximum Likelihood is based, as are many brilliant ideas, on an extremely simple concept. Because of its importance we shall proceed rather slowly, starting with a very simple example.

Example 13.1. Three urns each contain a large number of balls (statisticians have a fixation with balls in urns, light bulbs and Prussian Cavalry Officers being kicked to death). One contains 60%, one 70% and the other 80% of white balls.

An urn is chosen at random and we wish to determine which one it was. This is to be done on the basis of number of white balls observed in a sample of 7. We can either assume that the balls are taken with replacement or rely on the statement that there is a large number of balls in each urn to allow us to use the Binomial distribution.

Using the Binomial distribution we can find for each urn the probabilities of obtaining 0, 1, 2, … , 7 white balls.

Number of white balls	**Probability** $p = 0.6$	$p = 0.7$	$p = 0.8$
0	0.002	0.000	0.000
1	0.017	0.004	0.000
2	0.077	0.025	0.004
3	0.193	0.097	0.029
4	0.290	0.227	0.115
5	0.261	0.318	0.275
6	0.131	0.247	0.367
7	0.028	0.082	0.210

The emboldened elements in the table emphasize the largest probability in each row. An intuitively sensible solution to our problem would seem to be

1. Select the row corresponding to the number of white balls observed
2. Find the highest probability in a row

3. Opt for the proportion (urn) that heads the column containing the probability chosen above

So

1. If our sample contains 0, 1, 2, 3 or 4 white balls decide that the urn chosen was that containing 60% of white balls.

2. If our sample contains 5 white balls decide that the urn chosen was that containing 70% of white balls.

3. If our sample contains 6 or 7 white balls decide that the urn chosen was that containing 80% of white balls.

We note that in this example we very neatly circumvented Bayes' theorem because each urn was chosen at random and as a consequence each had a 1/3 chance of selection.
The example above could possibly occur but it is somewhat contrived in order to illustrate the principle of Maximum Likelihood. A more realistic situation would be to be presented with just one urn containing a large number of balls of which an unknown proportion was white.

Example 13.2. (The Binomial Distribution) We now have a single urn containing a large number of balls an unknown proportion of which are white. We take a sample of 7 balls and note that the number of white balls as 5. If we still do not like the artificiality of this example it can be redress as follows: we want to know what proportion of seed in a batch is fertile. We plant 7 and 5 germinate. We assume that all fertile seeds will germinate (an assumption that I know from bitter experience to be totally unwarranted). If the unknown proportion is p then the probability of obtaining the result observed is

$$P(5;7) = \binom{7}{5} p^5 (1-p)^2$$

If we regard this as a function of p then we can draw the graph as below.

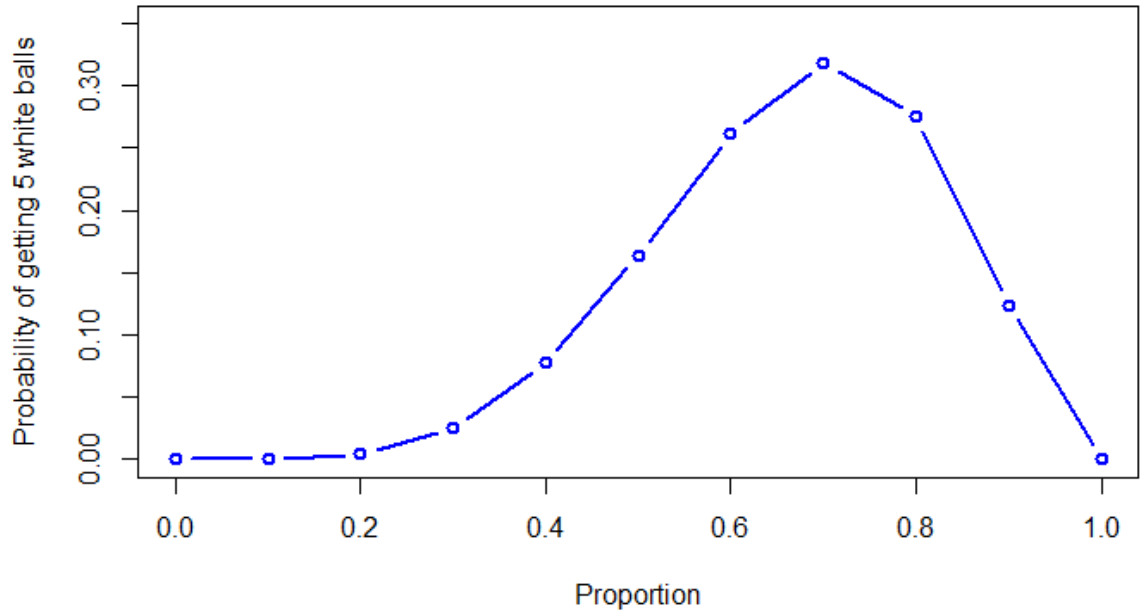


Figure 13.1.

As we see this function reaches a maximum when $p \approx 0.7$ and it would seem reasonable to take this value as our estimate of the unknown true proportion p.

13.3. Maximum Likelihood Estimation (MLE)

Let $X_1, X_2, \dots, X_n$ be the independent random variables taken from a discrete probability distribution $f(x;\theta)$ where θ is a single parameter of the distribution. Suppose that a sample of observations $X_1, X_2, \dots, X_n$ is taking values $x_1, x_2, \dots, x_n$.

$\mathbb{DF}$: (The Likelihood Function)

$$L = L(\theta|x_1, \dots, x_n) = f(x_1, \dots, x_n; \theta) = f(x_1;\theta) \times f(x_2;\theta) \times f(x_n;\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

is the joint distribution of the random variables $X_1, X_2, \dots, X_n$ which is called the likelihood function.

Note that L is not a function of x's; it is function of θ. When it is regarded as a function of θ, it is called the Likelihood Function and is denoted by $L(\theta|x_1, \dots, x_n)$.

We can note the following:

1. It is more usual to write the Likelihood Function as $L(\theta)$ or even L.
2. We would be wrong to think of $L(\theta)$ as a probability density function. Although $L(\theta)$ shares with probabilities the fact that its values can never be negative, it is a function of θ and not the x's. For example, its integral over the range of acceptable values of θ will almost certainly be 1.

$$\int_{\Omega} L(\theta) d\theta \neq 1$$

3. The parameter θ may be a vector $\boldsymbol{\theta}$ of parameters.

$\mathbb{DF}$: (Maximum Likelihood Estimate)

The Maximum Likelihood Estimate of $\boldsymbol{\theta} = (\theta_1, \dots, \theta_k)$ are the values $\theta_1, \dots, \theta_k$, denoted by $\hat{\theta}_1, \dots, \hat{\theta}_k$ which jointly maximize the Likelihood Function.

The following points may be noted:

1. In many cases the Likelihood Function will be continuously differentiable, and the range of the X's will not depend on any of θ's. When this occurs, we will be able to obtain the MLE's of the θ's by differentiation.
2. If we consider any monotone increasing function of L then its maximum will occur at the value of θ as that for the maximum of L. In particular, we will often find it convenient to consider ln $L(\theta)$ often abbreviated to ln L. This is because we can convert the products of the p.d.f.'s into sums of logarithms. Further using the chain rule of differentiation
$$\frac{d(\ln L)}{d\theta} = \frac{1}{L} \times \frac{dL}{d\theta}$$
which re-emphasizes that the stationary points of ln L occur at the same θ values as those of L.

3. If we have a $k-$ vector θ's to estimate, then we will need to solve simultaneously a set of k equations of the form
$$\frac{dL}{d\theta_j} = 0$$
or
$$\frac{d(\ln L)}{d\theta_j} = 0$$

Note that it is convenient to work with the natural logarithm of the likelihood function.

Example 13.3. (The Binomial distribution – continuing the previous example). We will generalize the previous to consider a sample of n balls of which r are white and we wish to determine the population proportion p of white balls. The Likelihood Function is

$$L(p) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$\ln L = \ln \binom{n}{r} + r \ln p + (n-r) \ln(1-p)$$

$$\frac{d(\ln L)}{dp} = 0 + \frac{r}{p} - \frac{n-r}{1-p} = 0$$

when $p = \hat{p}$ i.e

$$\frac{r}{\hat{p}} - \frac{n-r}{1-\hat{p}} = 0$$

$$\hat{p} = \frac{r}{n}$$

Substituting in the values from Example 13.2 of 7 and 5 for n and r we get the Maximum Likelihood estimate for p of $\hat{p} = 0.714$. It is somewhat ratifying that the MLE of p is the intuitive one, i.e. we estimate the population proportion with the sample proportion.

The example given above is a useful and clear example of the Maximum Likelihood estimation. However, it is not typical of most examples. This is because the Binomial distribution is itself derived from another distribution, the simpler Bernoulli distribution. We were saved the problem of multiplying p.d.f.'s to get a joint p.d.f. This is rectified in the example below.

Example 13.4. We wish to estimate the parameter k in the distribution

$$f(x;k) = (k+1)x^k \text{ for } 0 < x < 1 \text{ and } k > -1$$

using MLE. We also have a sample of 8 values

0.5	0.2	0.9	0.8	0.8	0.4	0.7	0.9

Solution

We will first consider a slightly more general case of n observations

$$L(k) = \prod_{i=1}^{n}(k+1)x_i^k = (k+1)^n \prod_{i=1}^{n} x_i^k$$

Therefore,

$$\ln L = n\ln(k+1) + k\ln\left(\prod_{i=1}^{n} x_i^k\right) = n\ln(k+1) + k\sum_{i=1}^{n}\ln x_i$$

$$\frac{d(\ln L)}{dk} = \frac{n}{k+1} + \sum_{i=1}^{n}\ln x_i = 0$$

when $k = \hat{k}$, the MLE of k.

So,

$$\frac{n}{\hat{k}+1} = -\sum_{i=1}^{n}\ln x_i = 0$$

$$\hat{k} = -1 - \frac{8}{-4.233} = 0.890$$

This can be compared the Method of Moments estimator which gave a value of $\hat{k} = 0.857$.

Note that the term $\frac{\sum \ln x}{n} = \frac{1}{n} \ln \prod x = \ln \sqrt[n]{x_1 \times x_2 \times \ldots \times x_n}$ is the logarithm of geometric mean.

13.4 The Invariance Principle

$\mathbb{TH}$: If $\hat{\theta}$ is the Maximum Likelihood Estimator of θ and $\tau(\theta)$ is a function of θ the Maximum Likelihood Estimator of $\tau(\theta)$ is

$$\widehat{\tau(\theta)} = \tau(\hat{\theta})$$

So having found, for example, the MLE of μ we can state that the MLE of $\ln(\mu)$ is $\ln(\bar{X})$.

Maximum likelihood estimators do not necessarily enjoy the property of unbiasedness. However, the maximum likelihood estimators do have very important asymptotic properties.

13.5 Properties of MLE's

Maximum Likelihood Estimators are not necessarily unbiased nor are they efficient. That is the bad news. The good news is that Maximum Likelihood Estimators enjoy an impressive array of asymptotic properties (i.e $n \to \infty$)..

1. MLE's are consistent estimators.
   - Put another way they are asymptotically unbiased.
   - In other words, as the sample size grows, the MLE converges to the true parameter.
2. MLE's are asymptotically normally distributed.
   - While they may be biased in small samples, the bias tends to disappear as n increases
3. MLE's are asymptotically efficient.
   - This means that asymptotically there is no more efficient estimator, i.e. one with the smaller variance.
   - There is a remarkable theorem that states what the absolute minimum variance is.
   - This minimum variance is called Cramér-Rao lower bound.

14 Statistical Formula and Statistical Tables

14.1 Statistical Formula

Expected value or mean

$$\mu = E[X] = \begin{cases} \sum_{all\ x} xf(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} xf(x), & \text{if } X \text{ is continuous} \end{cases}$$

Sums of Random Variables

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

$$\text{Var}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i^2 Var[X_i] + 2\underbrace{\sum\sum}_{i<j} a_i a_j \text{Cov}(X_i, X_j)$$

Variance

$$\text{Var}(X) = E[X^2] - \mu^2 = E[X^2] - \{E[X]\}^2$$

Moments

$$\mu_r' = E[X^r] = \begin{cases} \sum_{R} x^r f(x) & \text{if } X \text{ is discrete} \\ \int_{R} x^r f(x)dx & \text{if } X \text{ is continuous} \end{cases}$$

Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Moment Generating Function

$$M_X(t) = E[e^{tX}]$$

$$M_X(t) = \begin{cases} \sum_{all\ x} e^{tx} f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} e^{tx} f(x)dx & \text{if } X \text{ is continuous} \end{cases}$$

$$\mu_r' = \left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0}$$

Maximum likelihood estimator

$$L = L(\theta|x_1, \dots, x_n) = \prod_{i=1}^{n} f(x_i; \theta)$$

or

$$\ln(L) = \sum_{i=1}^{n} \ln(f(x_i; \theta))$$

Chi-square

$$X^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

The Mann-Whitney/Wilcoxon Rank sum test

Mann and Whitney modified the test and based it on

$$U_1 = W_1 - \frac{n_1(n_1 + 1)}{2}$$

or

$$U_2 = W_2 - \frac{n_2(n_2 + 1)}{2}$$

and

$$U = \min(U_1, U_2)$$

Mean and Variance

$$\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad\qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Two Independent Samples (Known variance or large samples):

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a $N(0,1)$ distribution

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

has approximately $N(0,1)$ distribution

Notes by Dr Jack Sutton (email: j.sutton@derby.ac.uk)

$$Z = \frac{P_1 - P_2 - (\pi_1 - \pi_2)}{\sqrt{\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}}}$$

has approximately $N(0,1)$ distribution

Small Samples

$$T = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a t-distribution with $v = n_1 + n_2 - 2$ df where

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

is a pooled estimate of a common variance σ^2.

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$$

has an F distribution with $(n_1 - 1, n_2 - 1)$ df

14.2 Common distributions

14.2.1 Discrete distributions

Name	$\boldsymbol{f(x)}$	$\boldsymbol{E[X]}$	**var[$\boldsymbol{X}$]**
Geometric if $X \sim Geom(p)$	$f(x) = p(1-p)^{x-1} = pq^{x-1}$	$E[X] = \frac{1}{p}$	$\text{var}(X) = \frac{1-p}{p^2}$
	for $x = 1,2,3,\ldots$ where p is the probability of success on each trial.		
Negative Binomial if $X \sim NegBin(r,p)$	$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r} = \binom{x-1}{r-1} p^r q^{x-r}$	$E[X] = \frac{r}{p}$	$\text{var}(X) = \frac{r(1-p)}{p^2}$
	for $x = r, r+1, r+2, \ldots$ where r is the number of successes to be achieved and p is the probability of success on each trial.		
Binomial if $X \sim Bin(n,p)$	$f(x) = \binom{n}{x} p^x (1-p)^{n-x} = \binom{n}{x} p^x q^{n-x}$	$E[X] = np$	$\text{var}(X) = np(1-p) = npq$
	for $x = 0,1,2,\ldots,n$ where n is the total number of trials and p is the probability of success on each individual trial.		

Poisson if $X \sim Po(\lambda)$	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$	$E[X] = \lambda$	$\text{var}(X) = \lambda$
	for $x = 0,1,2,\ldots$ where λ is the average rate of events per interval.		

14.2.2 Continuous distributions

Name	$\boldsymbol{f(x)}$	$\boldsymbol{E[X]}$	**var[$\boldsymbol{X}$]**
Uniform if $X \sim U(a,b)$	$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{elsewhere} \end{cases}$	$E[X] = \frac{a+b}{2}$	$\text{var}[X] = \frac{(b-a)^2}{12}$
	where a and b represent the lower and upper bounds of the distribution.		
Exponential if $X \sim Exp(\lambda)$	$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & elsewhere \end{cases}$	$E[X] = \frac{1}{\lambda}$	$\text{var}[X] = \frac{1}{\lambda^2}$
	where λ is the rate parameter, which is the average number of events per unit of time.		
Normal if $X \sim N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$	$E[X] = \mu$	$\text{var}[X] = \sigma^2$
	where μ is the mean and σ^2 is the variance parameters.		

14.3 Tables

14.3.1 Standard Normal

Statistical Tables

Standard Normal Probabilities
(The table is based on the area **P** under the standard normal probability curve, below the respective **z**-statistic.)



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-4.0	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00002	0.00002	0.00002	0.00002
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
-2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
-2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
-2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
-2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
-2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
-2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
-2.3	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
-2.2	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
-2.1	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
-2.0	0.02275	0.02222	0.02169	0.02118	0.02067	0.02018	0.01970	0.01923	0.01876	0.01831
-1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
-1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
-1.7	0.04456	0.04363	0.04272	0.04181	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
-1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
-1.5	0.06681	0.06552	0.06425	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07214	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
-1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09852
-1.1	0.13566	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.0	0.15865	0.15625	0.15386	0.15150	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.9	0.18406	0.18141	0.17878	0.17618	0.17361	0.17105	0.16853	0.16602	0.16354	0.16109

-0.8	0.21185	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.7	0.24196	0.23885	0.23576	0.23269	0.22965	0.22663	0.22363	0.22065	0.21769	0.21476
-0.6	0.27425	0.27093	0.26763	0.26434	0.26108	0.25784	0.25462	0.25143	0.24825	0.24509
-0.5	0.30853	0.30502	0.30153	0.29805	0.29460	0.29116	0.28774	0.28434	0.28095	0.27759
-0.4	0.34457	0.34090	0.33724	0.33359	0.32997	0.32635	0.32276	0.31917	0.31561	0.31206
-0.3	0.38209	0.37828	0.37448	0.37070	0.36692	0.36317	0.35942	0.35569	0.35197	0.34826
-0.2	0.42074	0.41683	0.41293	0.40904	0.40516	0.40129	0.39743	0.39358	0.38974	0.38590
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43250	0.42857	0.42465
-0.0	0.50000	0.49601	0.49202	0.48803	0.48404	0.48006	0.47607	0.47209	0.46811	0.46414

14.3.2 Binomial

Binomial probabilities $b(x;n,p)=\binom{n}{x}p^x q^{n-x} \quad x=0,1,2,...,n$

		p										
	x	0.01	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
n=14	0	0.8687	0.4877	0.2288	0.1028	0.0440	0.0178	0.0068	0.0024	0.0008	0.0002	0.0001
	1	0.1229	0.3593	0.3559	0.2539	0.1539	0.0832	0.0407	0.0181	0.0073	0.0027	0.0009
	2	0.0081	0.1229	0.2570	0.2912	0.2501	0.1802	0.1134	0.0634	0.0317	0.0141	0.0056
	3	0.0003	0.0259	0.1142	0.2056	0.2501	0.2402	0.1943	0.1366	0.0845	0.0462	0.0222
	4		0.0037	0.0349	0.0998	0.1720	0.2202	0.2290	0.2022	0.1549	0.1040	0.0611
	5		0.0004	0.0078	0.0352	0.0860	0.1468	0.1963	0.2178	0.2066	0.1701	0.1222
	6			0.0013	0.0093	0.0322	0.0734	0.1262	0.1759	0.2066	0.2088	0.1833
	7			0.0002	0.0019	0.0092	0.0280	0.0618	0.1082	0.1574	0.1952	0.2095
	8				0.0003	0.0020	0.0082	0.0232	0.0510	0.0918	0.1398	0.1833
	9					0.0003	0.0018	0.0066	0.0183	0.0408	0.0762	0.1222
	10						0.0003	0.0014	0.0049	0.0136	0.0312	0.0611
	11							0.0002	0.0010	0.0033	0.0093	0.0222
	12								0.0001	0.0005	0.0019	0.0056
	13									0.0001	0.0002	0.0009
	14											0.0001
n=15	0	0.8601	0.4633	0.2059	0.0874	0.0352	0.0134	0.0047	0.0016	0.0005	0.0001	
	1	0.1303	0.3658	0.3432	0.2312	0.1319	0.0668	0.0305	0.0126	0.0047	0.0016	0.0005
	2	0.0092	0.1348	0.2669	0.2856	0.2309	0.1559	0.0916	0.0476	0.0219	0.0090	0.0032
	3	0.0004	0.0307	0.1285	0.2184	0.2501	0.2252	0.1700	0.1110	0.0634	0.0318	0.0139
	4		0.0049	0.0428	0.1156	0.1876	0.2252	0.2186	0.1792	0.1268	0.0780	0.0417
	5		0.0006	0.0105	0.0449	0.1032	0.1651	0.2061	0.2123	0.1859	0.1404	0.0916
	6			0.0019	0.0132	0.0430	0.0917	0.1472	0.1906	0.2066	0.1914	0.1527
	7			0.0003	0.0030	0.0138	0.0393	0.0811	0.1319	0.1771	0.2013	0.1964
	8				0.0005	0.0035	0.0131	0.0348	0.0710	0.1181	0.1647	0.1964
	9				0.0001	0.0007	0.0034	0.0116	0.0298	0.0612	0.1048	0.1527
	10					0.0001	0.0007	0.0030	0.0096	0.0245	0.0515	0.0916
	11						0.0001	0.0006	0.0024	0.0074	0.0191	0.0417
	12							0.0001	0.0004	0.0016	0.0052	0.0139
	13								0.0001	0.0003	0.0010	0.0032
	14										0.0001	0.0005
	15											

n=19	0	0.8262	0.3774	0.1351	0.0456	0.0144	0.0042	0.0011	0.0003	0.0001		
	1	0.1586	0.3774	0.2852	0.1529	0.0685	0.0268	0.0093	0.0029	0.0008	0.0002	
	2	0.0144	0.1787	0.2852	0.2428	0.1540	0.0803	0.0358	0.0138	0.0046	0.0013	0.0003
	3	0.0008	0.0533	0.1796	0.2428	0.2182	0.1517	0.0869	0.0422	0.0175	0.0062	0.0018
	4		0.0112	0.0798	0.1714	0.2182	0.2023	0.1491	0.0909	0.0467	0.0203	0.0074
	5		0.0018	0.0266	0.0907	0.1636	0.2023	0.1916	0.1468	0.0933	0.0497	0.0222
	6		0.0002	0.0069	0.0374	0.0955	0.1574	0.1916	0.1844	0.1451	0.0949	0.0518
	7			0.0014	0.0122	0.0443	0.0974	0.1525	0.1844	0.1797	0.1443	0.0961
	8			0.0002	0.0032	0.0166	0.0487	0.0981	0.1489	0.1797	0.1771	0.1442
	9				0.0007	0.0051	0.0198	0.0514	0.0980	0.1464	0.1771	0.1762
	10				0.0001	0.0013	0.0066	0.0220	0.0528	0.0976	0.1449	0.1762
	11					0.0003	0.0018	0.0077	0.0233	0.0532	0.0970	0.1442
	12						0.0004	0.0022	0.0083	0.0237	0.0529	0.0961
	13						0.0001	0.0005	0.0024	0.0085	0.0233	0.0518
	14							0.0001	0.0006	0.0024	0.0082	0.0222
	15								0.0001	0.0005	0.0022	0.0074
	16									0.0001	0.0005	0.0018
	17										0.0001	0.0003
	18											
	19											

14.3.3 Wilcoxon Signed Ranks Test

n	Two-Tailed Test		One-Tailed Test	
	$\alpha = .05$	$\alpha = .01$	$\alpha = .05$	$\alpha = .01$
5	--	--	0	--
6	0	--	2	--
7	2	--	3	0
8	3	0	5	1
9	5	1	8	3
10	8	3	10	5
11	10	5	13	7
12	13	7	17	9
13	17	9	21	12
14	21	12	25	15
15	25	15	30	19
16	29	19	35	23
17	34	23	41	27
18	40	27	47	32
19	46	32	53	37
20	52	37	60	43
21	58	42	67	49
22	65	48	75	55
23	73	54	83	62
24	81	61	91	69
25	89	68	100	76
26	98	75	110	84
27	107	83	119	92
28	116	91	130	101
29	126	100	140	110
30	137	109	151	120

14.3.4 Mann-Whitney/Wilcoxon rank-sum two sample test

5%

critical region: $U \leqslant$ tabulated value

n_L \ n_S	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
2	-	-	-	-	-	-	0	0	0	0	1	1	1	1	1	2	2	2	2	3	3	3	3	3
3	-	-	-	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9	9	10	10
4	-	-	-	1	2	3	4	4	5	6	7	8	9	10	11	11	12	13	14	15	16	17	17	18
5	-	-	-	-	3	5	6	7	8	9	11	12	13	14	15	17	18	19	20	22	23	24	25	27
6	-	-	-	1	3	6	8	10	11	13	14	16	17	19	21	22	24	25	27	29	30	32	33	35
7	-	-	0	1	3	6	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44
8	-	-	0	2	4	6	10	15	17	19	22	24	26	29	31	34	36	38	41	43	45	48	50	53
9	-	-	1	3	5	7	9	15	20	23	26	28	31	34	37	39	42	45	48	50	53	56	59	62
10	-	-	0	2	4	6	9	11	13	26	29	33	36	39	42	45	48	52	55	58	61	64	67	71
11	-	-	0	2	5	7	10	13	16	18	33	37	40	44	47	51	55	58	62	65	69	73	76	80
12	-	-	1	3	6	9	12	15	18	21	24	41	45	49	53	57	61	65	69	73	77	81	85	89
13	-	-	1	3	7	10	13	17	20	24	27	31	50	54	59	63	67	72	76	80	85	89	94	98
14	-	-	1	4	7	11	15	18	22	26	30	34	38	59	64	69	74	78	83	88	93	98	102	107
15	-	-	2	5	8	12	16	20	24	29	33	37	42	46	70	75	80	85	90	96	101	106	111	117
16	-	-	2	5	9	13	18	22	27	31	36	41	45	50	55	81	86	92	98	103	109	115	120	126
17	-	-	2	6	10	15	19	24	29	34	39	44	49	54	60	65	93	99	105	111	117	123	129	135
18	-	-	2	6	11	16	21	26	31	37	42	47	53	58	64	70	75	106	112	119	125	132	138	145
19	0	3	7	12	17	22	28	33	39	45	51	57	63	69	74	81	87	92	119	126	133	140	147	154
20	0	3	8	13	18	24	30	36	42	48	54	60	67	73	79	86	92	99	134	141	149	156	163	170
21	0	3	8	14	19	25	32	38	44	51	58	64	71	78	84	91	98	105	112	150	157	165	173	181
22	0	4	9	14	21	27	34	40	47	54	61	68	75	82	89	96	104	111	118	125	166	174	182	190
23	0	4	9	15	22	29	35	43	50	57	64	72	79	87	94	102	109	117	125	132	140	183	192	200
24	0	4	10	16	23	30	37	45	52	60	68	75	83	91	99	107	115	123	131	139	147	155	201	211
25	0	5	10	17	24	32	39	47	55	63	71	79	87	96	104	112	121	129	138	146	155	163	172	221

1%

equal sample sizes

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
5%	-	-	-	0	2	5	8	13	17	23	30	37	45	55	64	75	87	99	113	127	142	158	175	192	211
1%	-	-	-	-	0	2	4	7	11	16	21	27	34	42	51	60	70	81	93	105	118	133	148	164	180

n	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
5%	230	250	272	294	317	341	365	391	418	445	473	503	533	564	596	628	662	697	732	769	806	845	884	924	965
1%	198	216	235	255	276	298	321	344	369	394	420	447	475	504	533	564	595	627	660	694	729	765	802	839	877

12.3.5 The Chi-square distribution

The Chi-square distribution

Rejection zone

df	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	---	---	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401

df	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40. 289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

12.3.6 Tukey's Wholly Significant Difference

Table 11

TUKEY'S WHOLLY SIGNIFICANT DIFFERENCE (5% LEVEL)

The table gives the values of the Wholly Significant Difference/s $(\bar{x})$ where:

W.S.D. = 'Wholly Significant Difference' between means at 5% level.
s = Estimate of standard deviation
$s(\bar{x})$ = Estimate of standard error of the means = $s/\sqrt{n}$
k = Number of means under test
ν = Degrees of freedom associated with s^2, the estimate of error variance
n = Number of readings in each mean

ν \ k	2	3	4	5	6	7	8	9	10	12	15	20	30	60
1	21.96	28.80	34.56	39.60	44.28	48.78	53.10	55.62	57.96	62.46	69.12	77.58	91.98	116.64
2	6.83	8.54	9.88	10.98	11.95	12.87	13.73	14.27	14.76	15.68	17.02	18.67	21.41	26.11
3	4.89	6.00	6.84	7.50	8.07	8.60	9.08	9.41	9.69	10.22	10.98	11.89	13.40	15.96
4	4.19	5.10	5.76	6.27	6.70	7.10	7.45	7.70	7.92	8.31	8.88	9.55	10.62	12.47
5	3.86	4.66	5.24	5.68	6.04	6.37	6.66	6.88	7.06	7.39	7.86	8.41	9.28	10.77
6	3.64	4.38	4.91	5.31	5.63	5.92	6.17	6.37	6.53	6.82	7.23	7.70	8.45	9.73
7	3.50	4.20	4.70	5.06	5.35	5.62	5.85	6.03	6.17	6.44	6.81	7.24	7.91	9.05
8	3.41	4.08	4.55	4.89	5.17	5.41	5.62	5.79	6.00	6.18	6.53	6.92	7.53	8.57
9	3.34	3.98	4.44	4.76	5.03	5.26	5.46	5.62	5.75	5.98	6.31	6.68	7.25	7.99
10	3.28	3.91	4.35	4.66	4.91	5.13	5.32	5.48	5.61	5.83	6.14	6.49	7.02	7.94
12	3.19	3.80	4.22	4.52	4.75	4.96	5.13	5.28	5.40	5.61	5.90	6.22	6.70	7.54
15	3.11	3.69	4.09	4.37	4.60	4.79	4.95	5.09	5.20	5.39	5.66	5.95	6.39	7.14
20	3.04	3.60	3.98	4.25	4.45	4.63	4.78	4.91	5.02	5.19	5.44	5.71	6.11	6.79
30	2.97	3.51	3.87	4.12	4.32	4.48	4.61	4.74	4.84	5.00	5.23	5.47	5.83	6.44
60	2.90	3.41	3.76	4.00	4.18	4.33	4.45	4.57	4.66	4.81	5.02	5.24	5.56	6.09
∞	2.82	3.32	3.66	3.88	4.04	4.18	4.29	4.40	4.49	4.63	4.82	5.01	5.29	5.76

See page 53 for an example on the use of this table.

Applied Statistics for Engineers, page 179. W. Volk (McGraw-Hill)
(Modification of tables in ***The Problem of Multiple Comparison***, mimeographed publication 1953, J. W. Tukey)